#6 · 8.2% of marks

Ch 14 — Mergers, Acquisitions & Corporate Restructuring

SM TYK theory Qs: **synergy** (para 2) · **reverse merger / takeover by reverse bid** (6) · **equity carve-out vs spin-off** (7.2) · **horizontal vs vertical merger** (3). Numericals: swap ratio, post-merger EPS / MPS, true cost, restructuring.

1 Conceptual framework (SM 1)

- → ***Corporate restructuring*** *= expansion or contraction of operations, or changes in assets, financial or ownership structure.* ***Internal****: new plant, R&D, hiving off non-core, divestment, sell-offs, demerger.* ***External****: M&A, JVs, strategic alliances.*
- → ***Merger*** *= two entities unify into one (legal name in India:* ***amalgamation****).* ***Acquisition*** *= one buys out another; both may remain separate entities.* ***Consolidation*** *= both dissolve, new one created.*
- → *Halsbury: amalgamation = blending of undertakings, shareholders of each becoming substantially shareholders of the amalgamated company. Only one company + rights varied =* ***reconstruction / arrangement****.*
- → *Law:* ***Companies Act 2013, Ch XV (s 230–240)*** *— compromise, arrangements, amalgamations, CDR, demergers, fast-track mergers, cross-border mergers.* ***Takeovers regulated by SEBI****; also FEMA, IT Act (****s 72A*** *revival), Competition Act 2002, IDRA 1951.*
- → *India's first takeover attempt:* ***Swaraj Paul – Escorts****.*

2 Rationale for M&A (SM 2) — TYK Q1 "synergy"

Synergistic operating economics: V(AB) > V(A) + V(B)*. Synergy = increase in performance of the combined firm over what the two would achieve independently (Mark Sirower, "The Synergy Trap"). Sources:* ***complementary activities*** *(one has branch network, other efficient production) and* ***economies of scale*** *(shared central services). Economies may be* ***real*** *(less input per unit) or* ***pecuniary*** *(lower prices for bulk inputs).*

- → ***Diversification*** *— unrelated merger cuts business risk → lower discount rate → higher value; more so with* ***independent / negatively correlated*** *income streams*
- → ***Taxation*** *— set off / carry forward of losses; target's losses set off against acquirer's profits*
- → ***Growth*** *— faster than organic: shorter "time to market" (no delays on site, plant, hiring)*
- → ***Consolidation of production capacity & market power*** *— reduced competition*

SM examples: growth / snuff competition — Airtel–Loop, Facebook–WhatsApp · capability — Google–Motorola (patents), Flipkart–Myntra · new market — Airtel–Zain, Cargill–Wipro · funds — Ranbaxy–Sun, Jaypee–UltraTech · tax — Burger King–Tim Hortons, DPL–WBPDCL.

③ Forms (types) of mergers (SM 3) — TYK Q4

Type	SM meaning
Horizontal	Same industry; larger market share; may move towards monopoly / near-monopoly to avoid competition
Vertical	Companies with a **buyer-seller** (or potential buyer-seller) relationship
Conglomerate	Unrelated businesses — neither horizontal nor vertical; pure = no common factors; purpose: use of financial resources, **enlarged debt capacity**, managerial synergy
Congeneric	Related through basic technologies, production processes or markets; extension of product-line / markets / technology; outward move within the industry
Reverse	Private company acquires a **public (shell)** company — bypasses the lengthy process of going public
Acquisition	Purchase of controlling interest via: (i) agreement with majority holder (ii) new shares by private agreement (iii) open-market purchase (open offer) (iv) cash / share issue (v) buyout offer to general body. Acquirer may merge or run target independently. Usually friendly.

Slump sale: business sold as a going concern for a lump sum without values to individual assets; transfer without allotting shares need not go through court (transferor continues).

④ Gains from merger (SM 4.1)

Value of acquirer + stand-alone value of target + **value of synergy** = combined value

Cost of acquisition = premium over market value + integration costs · **Net gain = synergy − premium**

SM eg: V_A 100, V_B 50, V_{AB} 175 → synergy **25**; premium 10 → net gain **15**

Value can be made from **non-synergistic** acquisitions too — **operating improvements** are a big source; managerial talent is the key; tie executive pay / equity to post-merger performance (Anslinger & Copeland, HBR 1996).

⑤ Scheme, evaluation, finance (SM 4.2–4.4)

- → **Scheme** = heart of the process; no prescribed form. Must vest **all assets & liabilities** of transferor in transferee (else dispose / settle first — otherwise defective, court won't sanction). Define the **effective date**. Courts won't re-examine valuation if fair — **Hindustan Lever**: SC won't interfere when > 99% shareholders approved.
- → **Financial evaluation** asks: (a) **maximum price** to pay (b) principal **risk** areas (c) **cash-flow & balance-sheet** implications (d) best way to **structure** the deal
- → **Financing**: cash, stock or both (in Definitive Agreement). Pay in shares even with cash available — keep a **war chest** for future deals, or acquirer's shares are **overpriced**. LBOs: immediate funding by **bridge financing**, long-term funds later.

THEORY · MCQ

Hostile takeover: *acquirer's board approaches target's shareholders directly through a **public announcement (tender offer)** after the target's board rejects the offer.*

⑥ Takeover strategies (SM 5.1)

Strategy	SM meaning
Street sweep	*Accumulate a large number of target shares **before** making an open offer → target left with no choice*
Bear hug	*Acquirer **threatens** an open offer → target's board agrees to a settlement for change of control*
Strategic alliance	*Disarm the acquirer by offering a **partnership** instead of a buyout; acquirer asserts control from within and takes over*
Brand power	*Alliance with powerful brands to displace target's brands, then buy out the weakened company*

⑦ Defensive tactics (SM 5.2)

Tactic	SM meaning
Divestiture	*Target divests / spins off some businesses into an independent subsidiary → less attractive*
Crown jewels	*Divestiture tactic = selling the crown jewels; not allowed in UK once the deal is known*
Poison pill	*Make itself unattractive, e.g. issue substantial **convertible debentures** to existing shareholders, convertible when threatened → voting shares rise sharply*
Poison put	*Target issues **bonds** encouraging holders to cash in at higher prices → cash drain*
Greenmail	*Target's management offers the bidder an incentive (higher-than-market price for **its** shares) to drop the takeover*
White knight	*Target offers itself to a **friendly company** to escape the hostile one (keep management)*
White squire	*Like white knight, but shares sold to a company **not interested in takeover** → management keeps control*
Golden parachute	*Hefty compensation to managers ousted due to takeover → reduces acquirer's interest*
Pac-man	*Target makes a **counter-bid** for the acquirer*

Greenmail = target **pays the bidder** to go away. White knight = friendly party **takes over**. White squire = friendly party buys shares but **doesn't take over**. SM: hostile takeovers should be avoided as far as possible.

⑧ Reverse merger (SM 6) — TYK Q2

- *Smaller company gains control of a larger one; the **healthy, prosperous** company's undertaking is merged into the **sick, non-viable** one (sick = net worth eroded). Also "**takeover by reverse bid**" / "**back door listing**".*
- ***3 tests***: *(i) transferor's assets > transferee's (ii) equity issued by transferee > its original issued capital (iii) change of control in transferee through a minority holder / group*
- ***Benefits***: *easy access to capital market · visibility · tax benefit on carry-forward losses of the acquired public company · cheaper, easier route to become public*

⑨ Divestiture (SM 7)

Reasons (4): (i) focus on core (ii) division not contributing enough revenue (iii) firm too big to handle (iv) cash needed urgently for other opportunities. ***Seller's perspective***: selling needs the same skill as buying; know your worth. Sell-side imperatives: competitor pressure; sale inevitable (no new technology, entry barriers, badly positioned, critical mass not reached, poor use of distribution, no new SBUs, not enough capital) · ***window of opportunity*** · focus on core competencies · best interest of shareholders.

Form (SM 7.2)	SM meaning
Sell-off / partial sell-off	Sale of asset, factory, division, product line or subsidiary for cash / securities; unit doesn't fit core strategy; market may undervalue combined business; raises cash (raiders sold subsidiaries to service LBO debt)
Spin-off	Part separated into a new firm; existing shareholders get ***proportionate ownership***; management parted; ***no fresh cash***. Reasons: separate identity · avoid takeover (valuable division spun off) · separate regulated & unregulated lines. Eg Future Group → Ezone
Split-up	Entire firm broken into a series of spin-offs; ***parent ceases to exist***; smaller units more manageable
Equity carve-out	Like spin-off but some shares sold to the public by ***IPO → brings cash***; parent keeps controlling stake; used when the subsidiary grows faster / has higher valuation; fails if carved out as a burden
Demerger of family business	~80% of Indian private companies family-managed; hive off unprofitable units for succession; step 1 identify core / non-core, step 2 cut interest through debt restructuring + sale of surplus assets

⚠ TRAP

TYK Q3 — carve-out vs spin-off: both create a separate company; spin-off shares go to ***existing shareholders*** pro rata with ***no cash***; carve-out sells shares to the ***public (IPO)***, ***raises cash***, parent retains control.

⑩ Financial restructuring (SM 8) = internal reconstruction

- → Internal changes in assets & liabilities ***with stakeholders' consent***, for firms with sizeable losses / eroded or negative net worth → fresh balance sheet free of losses and fictitious assets
- → Sacrifice order: ***equity shareholders most***, then preference holders, debenture holders, lenders, creditors — waivers, lower coupons, ***debt-to-equity conversion***
- → Results: (i) reduction / waiver of claims (ii) assets at real worth by revaluation (iii) gains on appreciation used to write off losses, fictitious assets, provide for doubtful debts

SM Illus 1 (XYZ): benefits = equity reduced 450 + preference 100 + debenture interest waived 26 + creditors 25% waived 75 + land revalued 50 = ***701***; used for P&L 485 + issue cost 5 + preliminary 10 + bad-debt provision 5 + P&M write-down 196 = ***701***. New BS total ***₹825 lakh***; cash = 130 + 150 − 74 = 206.

11 Ownership restructuring (SM 9)

- → **Going private**: listed company converted into private by buying back all outstanding shares (Essar Energy delisted from LSE, 2014); via management buyout or tender offer; usually **heavy debt**
- → **MBO**: company bought by its own management; useful to exit non-core divisions
- → **LBO**: financed entirely or partially **(50% or more) by borrowed funds**; target goes private; secured on acquired firm's physical assets; improve efficiency → cash flow repays debt → may go public again. Targets: market leaders, proven demand, strong management, steady growth. India's first: **Tata Tea – Tetley (2000)**, ₹2,135 cr, ~77% debt; Tata Steel – Corus (7.6 bn)
- → **Equity buyback**: surplus cash buys shares from public (opposite of IPO); shares cease to exist; raises promoter stake

Effects of buyback (SM): (i) controlling holders' % rises (ii) **EPS rises** → price steps up (iii) BS: assets (cash) and equity both fall; cash outflow under **financing activities** and in statement of changes in equity (iv) **ROA, ROE improve**. SM eg: ₹1.5 cr of ₹2 cr cash, 1 L shares bought → assets 5 → 3.5 cr, ROA **4%** → **5.71%**, EPS **2.00** → **2.22**.

12 Unlocking value (SM 10)

Horizontal → optimum size, market share, curb competition, use idle capacity · vertical → save costs, avoid taxes / duties · diversification · use idle funds of another company · merge export / trading co. with industrial co. (cash flow) · merge subsidiary into holding co. (cash flow) · take over a **shell** with licences · nourish a **sick unit** of the group (image). Restructuring → more competitive, survive adverse climate, new direction.

13 Premium & discount (SM 11)

Premiums: **market share, controlling stake, brand value**; discounts: **small player, unlisted company**. Work out scenarios with probabilities. **Timing** matters — economic cycle (GDP), stock market (multiples), global events. Synergy realisation risk (eg Satyam Infoway – India World, 2000). Use several methods; NAV / past-earnings methods may be inadequate for growing or intangible-heavy businesses.

14 M&A failures (SM 12)

5 steps of a successful programme

1 Manage pre-acquisition (keep **secrecy**) · 2 Screen candidates · 3 Eliminate misfits, value the rest · 4 Negotiate · 5 Post-merger integration

Why mergers fail

Acquirers **overpay** · synergy **over-estimated** · poor **post-merger integration** · **psychological** barriers. Evidence: Jensen & Ruback — target holders **~20%**, acquirer **~0%**; McKinsey — 61% of 116 failed, 23% succeeded

To succeed: decide post-merger tasks · choose managers from both companies (and outside) · set performance yardsticks and evaluate · motivate them.

15 Acquisition through shares (SM 13)

- → **Steps**: value acquirer's equity → value target's equity → maximum shares that can be exchanged → pessimistic & optimistic scenarios
- → **Exchange ratio** = acquirer's shares given per target share. SM eg: B's 1,00,000 shares at ₹230; A values itself ₹200 → 230/200 = **1.15** → 1,15,000 shares
- → Share count set by: (1) acquirer compares value per share with / without merger (2) seller compares its value with value of shares received (3) negotiation — reflects **bargaining position**. Fewer shares issued → acquirer better off, target worse off; both bear **synergy risk**
- → **Impact of P/E**: P/E = reciprocal of cost of equity, reflects risk; combined P/E may differ. Value = E × P/E → a **lower combined P/E can offset synergy**; a higher one can add value even without synergy

SM illustration	Result
Illus 2: X 3 L sh, MP 30, EPS 4; Y 2 L sh, MP 20, EPS 2.25	(a) EPS ratio 2.25/4 → 1,12,500 sh; EPS_{XY} = 16.5 L ÷ 4,12,500 = **₹4** (both unchanged) (b) 0.5 : 1 → 4 L sh; EPS **₹4.125**: X +0.125; Y equivalent 4.125 × 0.5 = 2.0625 (−0.1875)
Illus 3: A ₹10 L, 2 L sh, MP 75; B ₹3 L, 50,000 sh, MP 60	(i) MP basis: 50,000 × 60/75 = 40,000 sh (4 for 5); EPS = 13 L ÷ 2.4 L = **₹5.42** (ii) B's EPS 6 unchanged: 3 L ÷ 5 = 60,000 sh → **6 for 5**
Illus 4: Simpson (V 20 L, 50,000 sh) + Wilson (V 10 L); combined 35 L	20,000 sh → Wilson 2/7 = ₹10 L · 25,000 → 1/3 = ₹11.67 L · 30,000 → 3/8 = ₹13.125 L; Simpson gets the rest (25 / 23.33 / 21.875 L)

16 Exam working (ICAI answer patterns; app-written method)

Swap ratio = target's measure ÷ acquirer's (EPS / BV / MP / intrinsic), or weighted average. Banks: CRAR (higher better), NPA (lower better → acquirer ÷ target). **MTP Nov 2026**: 4.0 × 0.40 + 0.8 × 0.25 + 2.0 × 0.35 = **2.5**; promoters 34.50 of 57.50 L = **60%**; EPS 400 ÷ 57.50 = **₹6.956**.

EPS_{AB} = (E_A + E_B + synergy) ÷ (N_A + N_B × ER) · **MPS_{AB}** = EPS_{AB} × P/E_{AB} · **equivalent EPS** of target = EPS_{AB} × ER

Max ER (acquirer not diluted): EPS_{AB} = EPS_A · **Min ER** (target not worse): EPS_{AB} × ER = EPS_B (Illus 3(ii)); same on MPS basis

True cost, share deal = target's % of merged firm × V_{AB} − V_B. PYQ May 2024: 20% × 2,800 − 500 = **₹60 L**; cash deal = cash − V_B.

THEORY + LINKS

17 Cross-border M&A (SM 14)

Route for global growth; Tata Steel CFO: organic growth in India too slow for large capital projects, few domestic targets. **Motivating factors (5)**: globalisation of production & distribution · integration of global economies · expansion of international trade & investment · countries reforming laws and giving **investment & tax incentives** · privatisation of state enterprises and banking consolidation.

18 SPACs (SM 15)

- → "**Blank cheque company**": raises funds by IPO to acquire an **unidentified target within a specific time**; no operations / revenue; money in **escrow**; not acquired in time → delisted, money returned; holders may **redeem** if not interested; on approval target becomes public
- → **India**: no specific framework. Companies Act — RoC can strike off a company not commencing operations within **one year**; SPACs take **2–3 years** → enabling provisions needed. Not eligible for listing under SEBI; SEBI planning a framework. **IFSCA** (GIFT City) released a consultation paper (offer size, sponsor holding, minimum application / subscription)
- → **Advantages**: capital even in volatile markets · lower fees, faster listing · negotiate with a **single entity** not thousands of investors · skilled professionals pick the target
- → **Challenges**: complex accounting / reporting; public-company readiness in **3–5 months** of LOI; typically **more expensive than an IPO**; investors' money idle in trust up to two years

19 Correlation — how Ch 14 actually connects

Ch 14 idea	The actual link (app-written cross-reference; the Ch 14 side is SM)
Swap ratio on intrinsic value	**Ch 5 / Ch 13**: value each company first (DDM, FCFE at K_e, FCFF at WACC − debt, P/E × EPS), divide by shares, then ratio. A wrong valuation = wrong ratio.
P/E = 1 ÷ cost of equity	**Ch 5**: with zero growth $P = E \div K_e$, so $P/E = 1/K_e$. A riskier merged firm → higher K_e → lower P/E → can wipe out synergy (SM 13).
Diversification motive	**Ch 6**: negatively correlated income streams cut total risk ($r = -1$ case); but shareholders can diversify themselves — the gain must show up as a lower discount rate / higher debt capacity.
Net gain = synergy − premium	**Ch 3**: this is the acquirer's NPV — accept if positive; synergy is the PV of incremental cash flows.
Reverse merger	**FR Ind AS 103**: reverse acquisition — legal acquirer (listed shell) is the accounting acquiree; tested by who gets control.
Buyback effects	**FR Ind AS 7**: buyback outflow is a financing activity (SM says so); EPS rises because shares fall (Ind AS 33 weighted shares).
Cross-border deal	**Ch 10 / Ch 11**: price in foreign currency → exposure until settlement (hedge); value foreign cash flows via forecast rates or foreign discount rate.

Past papers

⑭ Mergers, Acquisitions and Corporate Restructuring — what ICAI asked (Sep 2025, Jan 2026, May 2026)

29 marks set across the last three attempts (8.2% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.

Attempt	Question	Marks	Concept tested
Sep 2025	MCQ CS-V 13-15	6	Acquisition by share exchange: EPS of both firms from P/E, post-acquisition EPS and post-acquisition P/E of acquirer (MCQ)
Jan 2026	MCQ CS-IV 10-12	6	Bank merger: swap ratio using weights on book value, market price, CRAR and NPA; shares issued; post-merger EPS (MCQ)
Jan 2026	Q1(b)	4	Complying with SEBI promoter-holding limit (84% to 75%) by bonus issue to minority holders at the same P/E: P/E, bonus ratio, post-bonus price and free-float market cap (Numerical)
May 2026	MCQ CS-III 7-9	6	Takeover defence and restructuring terms: poison pill, greenmail, carve-out (identify the tactic from a description) (MCQ)
May 2026	Q6(a)	7	Acquisition with cash, hybrid equity and hybrid convertible-bond offers: synergies, P/E, fair value per share, shares offered, ranking of the offers (Numerical)

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. Explain synergy in the context of Mergers and Acquisitions.

ICAI answer: "Please refer paragraph 2" – SM paragraph 2 reproduced

2. RATIONALE FOR MERGERS AND ACQUISITIONS

The most common reasons for Mergers and Acquisition (M&A) are:

- **Synergistic operating economics:** Synergy may be defined as follows:

  V (AB) >V(A) + V (B).

  In other words, the combined value of two firms or companies shall be more than their individual value. Synergy is the increase in performance of the combined firm over what the two firms are already expected or required to accomplish as independent firms (Mark L Sirower of Boston Consulting Group, in his book "The Synergy Trap"). This may be the result of complimentary services or economies of scale or both.

  A good example of complimentary activities can be a company having a good network of branches and other company having an efficient production system. Thus, the merged companies will be more efficient than individual companies.

On similar lines, economies of large scale is also one of the reasons for synergy benefits. The main reason is that, the large scale production results in lower average cost of production e.g. reduction in overhead costs on account of sharing of central services such as accounting and finances, office executives, top level management, legal, sales promotion and advertisement etc.

These economies can be "real" arising out of reduction in factor input per unit of output, whereas pecuniary economics are realized from paying lower prices for factor inputs for bulk transactions. Other factors for Synergies are as follows:

- **Diversification:** Merger between two unrelated companies would lead to reduction in business risk, which in turn will increase the market value consequent upon the reduction in discount rate/ required rate of return. Normally, greater the combination of statistically independent or negatively correlated income streams of merged companies, higher will be the reduction in business risk in comparison to companies having income streams which are positively correlated to each other.
- **Taxation:** The provisions of set off and carry forward of losses as per Income Tax Act may be another strong reason for the merger and acquisition. Thus, there will be Tax saving or reduction in tax liability of the merged firm. Similarly, in the case of acquisition the losses of the target company will be allowed to be set off against the profits of the acquiring company.
- **Growth:** Merger and acquisition mode enables the firm to grow at a rate faster than the other mode viz., organic growth. The reason being the shortening of 'Time to Market'. The acquiring company avoids delays associated with purchasing of building, site, setting up of the plant and hiring personnel etc.
- **Consolidation of Production Capacities and increasing market power:** Due to reduced competition, marketing power increases. Further, production capacity is increased by the combination of two or more plants. The following table shows the key rationale for some of the well-known transactions which took place in India in the recent past.

Rationale for M & A

Instantaneous growth, Snuffing out competition, Increased market share.	• Airtel – Loop Mobile (2014) (Airtel bags top spot in Mumbai Telecom Circle)
Acquisition of a competence or a capability	• Google – Motorola (2011) (Google got access to Motorola's 17,000 issued patents and 7500 applications)

Entry into new markets/product segments	• Airtel – Zain Telecom (2010) (Airtel enters 15 nations of African Continent in one shot)
Access to funds	• Ranbaxy – Sun Pharma (2014) (Daiichi Sankyo sold Ranbaxy to generate funds)
Tax benefits	• Burger King (US) – Tim Hortons (Canada) (2014) (Burger King could save taxes in future)
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Theory Q2

2. What is take over by reverse bid or Reverse Merger.

ICAI answer: "Please refer paragraph 6" – SM paragraph 6 reproduced

6. REVERSE MERGER

In ordinary cases, the company taken over is the smaller company; in a 'reverse takeover', a smaller company gains control of a larger one. The concept of takeover by reverse bid, or of reverse merger, is thus not the usual case of amalgamation of a sick unit which is non-viable with a healthy or prosperous unit but is a case whereby the entire undertaking of the healthy and prosperous company is to be merged and vested in the sick company which is non-viable. A company becomes a sick industrial company when there is erosion in its net worth. This alternative is also known as taking over by reverse bid.

The three tests to be fulfilled before an arrangement can be termed as a reverse takeover are specified as follows:

(i) the assets of the transferor company are greater than the transferee company,

(ii) equity capital to be issued by the transferee company pursuant to the acquisition exceeds its original issued capital, and

(iii) the change of control in the transferee company through the introduction of a minority holder or group of holders.

This type of merger is also known as 'back door listing'. This kind of merger has been started as an alternative to go for public issue without incurring huge expenses and passing through the cumbersome process. Thus, it can be said that reverse merger leads to the following benefits for the acquiring company:

- Easy access to capital market.
- Increase in visibility of the company in corporate world.
- Tax benefits on carry forward losses acquired (public) company.
- Cheaper and easier route to become a public company.

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(ii) equity capital to be issued by the transferee company pursuant to the acquisition exceeds its original issued capital, and

(iii) the change of control in the transferee company through the introduction of a minority holder or group of holders.

This type of merger is also known as 'back door listing'. This kind of merger has been started as an alternative to go for public issue without incurring huge expenses and passing through the cumbersome process. Thus, it can be said that reverse merger leads to the following benefits for the acquiring company:

- Easy access to capital market.
- Increase in visibility of the company in corporate world.
- Tax benefits on carry forward losses acquired (public) company.

- Cheaper and easier route to become a public company.

Theory Q3

3. What is an equity curve out? How does it differ from a spin off.

ICAI answer: "Please refer paragraph 7.2" – SM paragraph 7.2 reproduced

7.2 Different Forms

Different ways of divestment or demerger or divestitures are as follows:

7.2.1 Sell off / Partial Sell off

A sell off is the sale of an asset, factory, division, product line or subsidiary by one entity to another for a purchase consideration payable either in cash or in the form of securities. Partial Sell off, is a form of divestiture, wherein the firm sells its business unit or a subsidiary to another because it is deemed to be unfit with the company's core business strategy.

Normally, sell-offs are done because the subsidiary doesn't fit into the parent company's core strategy. The market may be undervaluing the combined businesses due to a lack of synergy between the parent and the subsidiary. So, the management and the board decide that the subsidiary is better off under a different ownership. Besides getting rid of an unwanted subsidiary, sell-offs also raise cash, which can be used to pay off debts. In the late 1980s and early 1990s, corporate raiders used debt to finance acquisitions. Then, after making a purchase they used to sell-off its subsidiaries to raise cash to service the debt. The raiders' method certainly makes sense if the sum of the parts is greater than the whole. When it isn't, deals are unsuccessful.

7.2.2 Spin-off

In this case, a part of the business is separated and created as a separate firm. The existing shareholders of the firm get proportionate ownership. So, there is no change in ownership and the same shareholders continue to own the newly created entity in the same proportion as previously in the original firm. The management of spin-off division is however, parted with. Spin-off does not bring fresh cash. The reasons for spin off may be:

(i) To give separate identity to a part/division.

(ii) To avoid the takeover attempt by a predator by making the firm unattractive to him since a valuable division is spun-off.

(iii) To create separate Regulated and unregulated lines of business.

Example: Kishore Biyani led Future Group spin off its consumer durables business, Ezone, into a separate entity in order to maximise value from it.

7.2.3 Split-up

This involves breaking up of the entire firm into a series of spin off (by creating separate legal entities). The parent firm no longer legally exists and only the newly created entities survive. For instance, a corporate firm has 4 divisions namely A, B, C, D. All these 4 divisions shall be split-up to create 4 new corporate firms with full autonomy and legal status. The original corporate firm is to be wound up. Since de-merged units are relatively smaller in size, they are logistically more convenient and manageable. Therefore, it is understood that spin-off and split-up are likely to enhance shareholders value and bring efficiency and effectiveness.

7.2.4 Equity Carve outs

This is like spin off, however, some shares of the new company are sold in the market by making a public offer, so this brings cash. More and more companies are using equity carve-outs to boost shareholder value. A parent firm makes a subsidiary public through an initial public offering (IPO) of shares, amounting to a partial sell-off. A new publicly listed company is created, but the parent keeps a controlling stake in the newly traded subsidiary.

A carve-out is a strategic avenue a parent firm may take when one of its subsidiaries is growing faster and carrying higher valuations than other businesses owned by the parent. A carve-out generates cash because shares in the subsidiary are sold to the public, but the issue also unlocks the value of the subsidiary unit and enhances the parent's shareholder value.

The new legal entity of a carve-out has a separate board, but in most carve-outs, the parent retains some control over it. In these cases, some portion of the parent firm's board of directors may be shared. Since the parent has a controlling stake, meaning that both firms have common shareholders, the connection between the two is likely to be strong. That said, sometimes companies carve-out a subsidiary not because it is doing well, but because it is a burden. Such an intention won't lead to a successful result, especially if a carved-out subsidiary is too loaded with debt or trouble, even when it was a part of the parent and lacks an established track record for growing revenues and profits.

7.2.5 Demerger or Division of Family-Managed Business

Around 80 per cent of private sector companies in India are family-managed companies. The family-owned companies are, under extraordinary pressure to yield control to professional managements, as, in the emerging scenario of a liberalised economy the capital markets are broadening, with attendant incentives for growth. So, many of these companies are arranging to hive off their unprofitable businesses or divisions with a view to meeting a variety of succession problems.

Even otherwise, a group of such family-managed companies may undertake restructuring of its operations with a view also to consolidating its core businesses. For this, the first step that may need to be taken is to identify core and non-core operations within the group. The second step may involve reducing interest burden through debt restructuring along with sale of surplus assets. The proceeds from the sale of assets may be employed for expanding by acquisitions and rejuvenation of its existing operations. The bottom line is that an acquisition must improve economies of scale, lower the cost of production, and generate and promote synergies. Besides acquisitions, therefore, the group may necessarily have to take steps to improve productivity of its existing operations.

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Theory Q4

4. Write a short note on Horizontal Merger and Vertical Merger.

ICAI answer: "Please refer paragraph 3" — SM paragraph 3 reproduced

3. FORMS (TYPES) OF MERGERS

A merger is generally understood to be a fusion of two companies. The term "merger" means the dissolution of one or more companies or firms or proprietorships to form or get absorbed into another company. By concept, merger increases the size of the undertakings. Following are the major types of mergers:

(i) **Horizontal Merger:** The two companies which have merged are in the same industry, normally the market share of the new consolidated company would be larger and it is possible that it may move closer to being a monopoly or a near monopoly to avoid competition.

(ii) **Vertical Merger:** This merger happens when two companies that have 'buyer-seller' relationship (or potential buyer-seller relationship) come together.

(iii) **Conglomerate Mergers:** Such mergers involve firms engaged in unrelated type of business operations. In other words, the business activities of acquirer and the target are neither related to each other horizontally (i.e., producing the same or competing products) nor vertically (having relationship of buyer and supplier).In a pure conglomerate merger, there are no important common factors between the companies in production, marketing, research and development and technology. There may however be some degree of overlapping in one or more of these common factors. Such mergers are in fact, unification of different kinds of businesses under one flagship company. The purpose of merger remains utilization of financial resources, enlarged debt capacity and also synergy of managerial functions.

(iv) **Congeneric Merger:** In these mergers, the acquirer and the target companies are related through basic technologies, production processes or markets. The acquired company represents an extension of product-line, market participants or technologies of the acquirer. These mergers represent an outward movement by the acquirer from its current business scenario to other related business activities within the overarching industry structure.

(v) **Reverse Merger:** Such mergers involve acquisition of a public (Shell Company) by a private company, as it helps the same private company to by-pass lengthy and complex process required to be followed in case it is interested in going public.

(vi) **Acquisition:** This refers to the purchase of controlling interest by one company in the share capital of an existing company. This may be by:

(i) an agreement with majority holder of Interest.

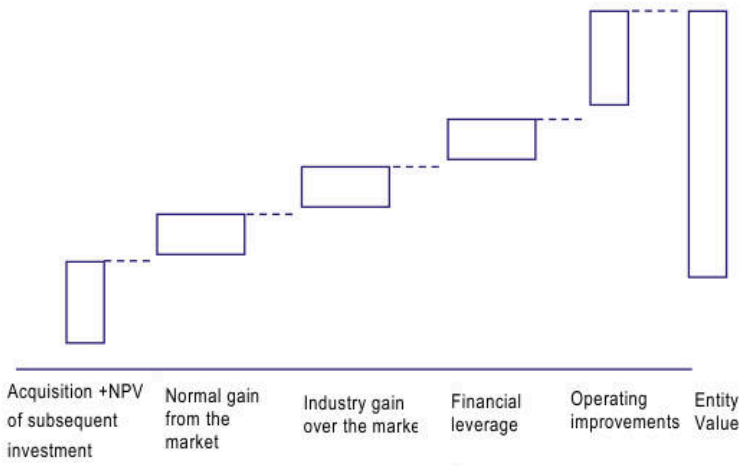
(ii) Purchase of new shares by private agreement.

(iii) Purchase of shares in open market (open offer)

(iv) Acquisition of share capital of a company by means of cash, issuance of shares.

(v) Making a buyout offer to general body of shareholders.

When a company is acquired by another company, the acquiring company has two choices, one, to merge both the companies into one and function as a single entity or two, to operate the taken-over company as an independent entity with changed management and policies. 'Merger' is the fusion of two independent firms on co-equal terms. 'Acquisition' is buying out of a company by another company and the acquired company usually loses its identity. Normally, this process is friendly.



Source: Patricial Anslinger and Thomas E Copeland, "Growth through Acquisitions : A Fresh look, Harvard Business Review Jan. – Feb -1996.

Acquisition of one of the businesses of a company, as a going concern by an agreement need not necessarily be routed through court, if the transfer of business is to be accomplished without allotting shares in the transferee company to the shareholders of the transferor company. This would tantamount to a simple acquisition. In this case the transferor company continues to exist and no change in shareholding is expected. If the sale takes place for a lumpsum consideration without attributing any individual values to any class of assets, such sales are called slump sales. The capital gains arising on slump sales were being exempt from income tax based on a decision of the Supreme Court of India.

ICAI answer: "Please refer paragraph 3" – SM paragraph 3 reproduced

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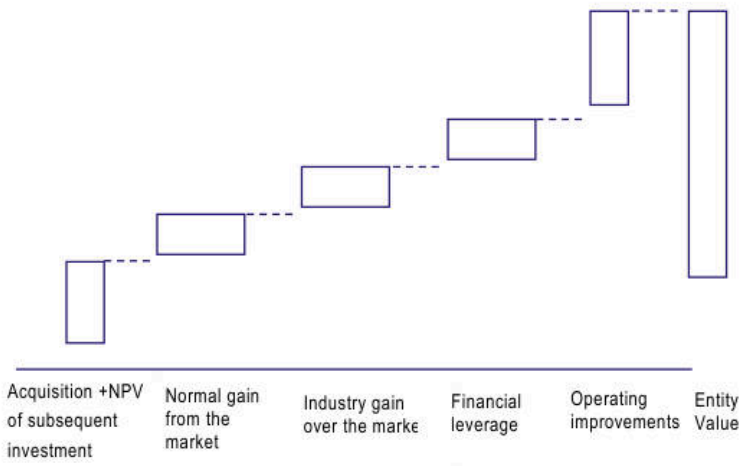
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① SM illustrations with ICAI solutions

8. Financial Restructuring

SM Ch 14 · Illustration 1

Illustration 1

The following is the Balance-sheet of XYZ Ltd. as on March 31st, 2013.

(₹ in lakh)

Liabilities	Amount	Assets	Amount
6 lakh Equity Shares of ₹100/- each	600	Land & Building	200
2 Lakh 14% Preference shares of ₹100/- each	200	Plant & Machinery Furniture & Fixtures	300 50
13% Debentures	200	Inventory	150
Debenture Interest accrued and Payable	26	Sundry debtors	70
Loan from Bank	74	Cash at Bank	130
Trade Creditors	300	Preliminary Expenses	10
		Cost of Issue of debentures	5
		Profit & Loss A/c	485
	1,400		1,400

The Company did not perform well and has suffered sizable losses during the last few years. However, it is now felt that the company can be nursed back to health by proper financial restructuring and consequently the following scheme of reconstruction has been devised:

(i) Equity shares are to be reduced to ₹ 25/- per share, fully paid up;

(ii) Preference shares are to be reduced (with Dividend rate of 10%) to equal number of shares of ₹50 each, fully paid up.

(iii) Debenture holders have agreed to forego interest accrued to them. Beside this, they have agreed to accept new debentures carrying a coupon rate of 9%.

(iv) Trade creditors have agreed to forgo 25 per cent of their existing claim; for the balance sum they have agreed to convert their claims into equity shares of ₹ 25/- each.

(v) In order to make payment for bank loan and augment the working capital, the company issues 6 lakh equity shares at ₹ 25/- each; the entire sum is required to be paid on application. The existing shareholders have agreed to subscribe to the new issue.

(vi) While Land and Building is to be revalued at ₹ 250 lakh, Plant & Machinery is to be written down to ₹ 104 lakh. A provision amounting to ₹ 5 lakh is to be made for bad and doubtful debts.

You are required to show the impact of financial restructuring/re-construction. Also, prepare the new balance sheet assuming the scheme of re-construction is implemented in letter and spirit.

Solution — ICAI

Solution

Impact of Financial Restructuring

(i) Benefits to XYZ Ltd.

		₹ in lakhs
(a)	**Reduction of liabilities payable**	
	Reduction in Equity Share capital (6 lakh shares x ₹75 per share)	450
	Reduction in Preference Share capital (2 lakh shares x ₹50 per share)	100
	Waiver of outstanding Debenture Interest	26
	Waiver from Trade Creditors (₹300 lakhs x 0.25)	75
		651
(b)	Revaluation of Assets	
	Appreciation of Land and Building (₹250 lakhs - ₹200 lakhs)	50
		701

(ii) Amount of ₹ 701 lakhs utilized to write off losses, fictious assets and over- valued assets.

	₹ in lakhs
Writing off Profit and Loss account	485
Cost of issue of debentures	5
Preliminary expenses	10
Provision for bad and doubtful debts	5
Revaluation of Plant and Machinery (₹300 lakhs – ₹104 lakhs)	196
	701

Balance sheet of XYZ Ltd as at_____ (after re-construction)

(₹ in lakhs)

Liabilities	Amount	Assets		Amount
21 lakhs Equity Shares of ₹25/- each	525	Land & Building		250
2 lakhs 10% Preference shares of ₹50/- each	100	Plant & Machinery		104
9% Debentures	200	Furnitures & Fixtures		50
		Inventory		150
		Sundry debtors	70	
			-5	65
		Cash-at-Bank (Balancing figure)*		206
	825			825

*Opening Balance of ₹ 130/- lakhs + Sale proceeds from issue of new equity shares ₹ 150/- lakhs – Payment of bank loan of ₹ 74/- lakhs = ₹ 206 lakhs.

It is worth mentioning that financial restructuring is unique in nature and is company specific. It is carried out, in practice when all shareholders sacrifice and understand that the restructured firm (reflecting its true value of assets, capital and other significant financial para meters) can now be nursed back to health. This type of corporate restructuring helps in the revival of firms that otherwise would have faced closure/liquidation.

13. Acquisition Through Shares

SM Ch 14 · Illustration 2

Illustration 2

Company X is contemplating the purchase of Company Y, Company X has 3,00,000 shares having a market price of ₹ 30 per share, while Company Y has 2,00,000 shares selling at ₹ 20 per share. The EPS are ₹ 4.00 and ₹ 2.25 for Company X and Y respectively. Managements of both companies are discussing two alternative proposals for exchange of shares as indicated below:

(i) in proportion to the relative earnings per share of two companies.

(ii) 0.5 share of Company X for one share of Company Y (0.5:1).

You are required:

(i) to calculate the Earnings Per share (EPS) after merger under two alternatives; and

(ii) to show the impact of EPS for the shareholders of two companies under both the alternatives.

Solution — ICAI

Solution

Working Notes: Calculation of total earnings after merger

Particulars	Company X	Company Y	Total
Outstanding shares	3,00,000	2,00,000	
EPS (₹)	4	2.25	
Total earnings (₹)	12,00,000	4,50,000	16,50,000

(i) (a) Calculation of EPS when exchange ratio is in proportion to relative EPS of two companies

Company X		3,00,000
Company Y	2,00,000 x 2.25/4	1,12,500
Total number of shares after merger		4,12,500

Company X

EPS before merger	=	₹ 4
EPS after merger = ₹ 16,50,000/4,12,500 shares	=	₹ 4

Company Y

EPS before merger	=	₹ 2.25
EPS after merger = EPS of Merged Entity after merger x Share Exchange ratio on EPS basis = $₹ 4 \times \frac{2.25}{4}$	=	₹ 2.25

(b) Calculation of EPS when share exchange ratio is 0.5 : 1

Total earnings after merger = ₹ 16,50,000

Total number of shares after merger = 3,00,000 + (2,00,000 x 0.5) = 4,00,000 shares

EPS after merger = ₹ 16,50,000/4,00,000 = ₹ 4.125

(ii) Impact of merger on EPS for shareholders of Company X and Company Y

(a) Impact on Shareholders of Company X

	(₹)
EPS before merger	4.000
EPS after merger	4.125
Increase in EPS	0.125

(b) Impact on Shareholders of Company Y

	(₹)
Equivalent EPS before merger	2.2500
Equivalent EPS after merger	2.0625
Decrease in EPS	0.1875

SM Ch 14 · Illustration 3

Illustration 3

A Ltd. is studying the possible acquisition of B Ltd. by way of merger. The following data are available:

Firm	After-tax earnings	No. of equity shares	Market price per share
A Ltd.	₹ 10,00,000	2,00,000	₹ 75
B Ltd.	₹ 3,00,000	50,000	₹ 60

(i) If the merger goes through by exchange of equity shares and the exchange ratio is set according to the current market prices, what is the new earnings per share for A Ltd.

(ii) B Ltd. wants to be sure that its earning per share is not diminished by the merger. What exchange ratio is relevant to achieve the objective?

Solution — ICAI

Solution

(i) The current market price is the basis of exchange of equity shares, in the proposed merger, shareholders of B Ltd. will get only 40,000 shares in all or 4 shares of A Ltd. for every 5 shares held by them, i.e.,

$$\frac{50,000 \times 60}{75} = 40,000$$

The total number of shares in A Ltd. will then be 2,40,000 and, ignoring any synergistic effect, the profit will be ₹ 13,00,000.The new earning per share (EPS) of A Ltd. will be ₹ 5.42, i.e., ₹ 13,00,000/2,40,000.

(ii) The present earnings per share of B Ltd. is ₹6/- (₹ 3,00,000 ÷ 50,000) and that of A Ltd. is ₹5/-, i.e., ₹ 10,00,000 ÷ 2,00,000.If B Ltd. wants to ensure that, even after merger, the earning per share of its shareholders should remain unaffected, then the exchange ratio will be 6 shares for every 5 shares.

The total number of shares of A Ltd. that will produce ₹ 3,00,000 profit is 60,000, (3,00,000 ÷ 5), to be distributed among, shareholders of B Ltd., giving a ratio of 6 shares in A for 5 shares in B.

Proof:

The shareholders of B Ltd. will get in all 60,000 share for 50,000 shares. It means after merger, their earning per share will be ₹ 5/-, i.e. $\frac{₹13,00,000}{2,60,000}$.

In all they will get ₹ 3,00,000, i.e., 60,000 x 5, as before.

SM Ch 14 · Illustration 4

Illustration 4

Simpson Ltd. is considering a merger with Wilson Ltd. The data below are in the hands of both Board of Directors. The issue at hand is how many shares of Simpson should be exchanged for Wilson Ltd. Both boards are considering three possibilities 20,000, 25,000 and 30,000 shares. You are required to construct a table demonstrating the potential impact of each scheme on each set of shareholders:

		Simpson Ltd.	Wilson Ltd.	Combined Post merger Firm 'A'
1.	Current earnings per year	2,00,000	1,00,000	3,50,000
2.	Shares outstanding	50,000	10,000	?
3.	Earnings per share (₹) (1÷ 2)	4	10	?
4.	Price per share (₹)	40	100	?
5.	Price-earning ratio [4 ÷ 3]	10	10	10
6.	Value of firm (₹)	20,00,000	10,00,000	35,00,000
7.	Expected Annual growth rate in earnings in foreseeable future	0	0	0

Solution — ICAI

Solution

The following table demonstrates the potential impact of the three possible schemes, on each set of shareholders:-

Number of Simpson Ltd.'s shares issued to shareholders of Wilson Ltd.	Exchange ratio [(1)/10,000 shares of Wilson Ltd.]	Number of Simpson Ltd.'s shares outstanding after merger [50,000+(1)]	Fraction of Simpson Ltd. (Post merger) owned by Wilson Ltd.'s shareholders [(1)/(3)]	Value of shares owned by Wilson Ltd.'s shareholders [(4)x 35,00,000]	Fraction of Simpson Ltd. (combined Post-merger owned by Simpson Ltd.'s share-holders [50,000/(3)]	Value of shares owned by Simpson Ltd.'s shareholders [(6) x 35,00,000]
(1)	(2)	(3)	(4)	(5)	(6)	(7)
20,000	2	70,000	2/7	10,00,000	5/7	25,00,000
25,000	2.5	75,000	1/3	11,66,667	2/3	23,33,333
30,000	3	80,000	3/8	13,12,500	5/8	21,87,500

Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical Q1

1. B Ltd. is a highly successful company and wishes to expand by acquiring other firms. Its expected high growth in earnings and dividends is reflected in its PE ratio of 17. The Board of Directors of B Ltd. has been advised that if it were to take over firms with a lower PE ratio than it own, using a share-for-share exchange, then it could increase its reported earnings per share. C Ltd. has been suggested as a possible target for a takeover, which has a PE ratio of 10 and 1,00,000 shares in issue with a share price of ₹ 15. B Ltd. has 5,00,000 shares in issue with a share price of ₹ 12.

   Calculate the change in earnings per share of B Ltd. if it acquires the whole of C Ltd. by issuing shares at its market price of ₹12. Assume the price of B Ltd. shares remains constant.

Solution — ICAI

1.

Total market value of C Ltd is = 1,00,000 x ₹ 15	= ₹ 15,00,000
PE ratio (given)	= 10
Therefore, earnings	= ₹ 15,00,000 /10
	= ₹ 1,50,000
Total market value of B Ltd. is = 5,00,000 x ₹ 12	= ₹ 60,00,000
PE ratio (given)	= 17
Therefore, earnings	= ₹ 60,00,000/17
	= ₹ 3,52,941

The number of shares to be issued by B Ltd.

₹ 15,00,000 ÷ 12	= 1,25,000
Total number of shares of B Ltd	= 5,00,000 + 1,25,000 = 6,25,000
The EPS of the new firm is	= (₹ 3,52,941+₹1,50,000)/6,25,000
	= ₹ 0.80
The present EPS of B Ltd is	= ₹ 3,52,941 /5,00,000
	= ₹ 0.71

So the EPS affirm B will increase from Re. 0.71 to ₹ 0.80 as a result of merger

Practical Q2

2. Elrond Limited plans to acquire Doom Limited. The relevant financial details of the two firms prior to the merger announcement are:

	Elrond Limited	Doom Limited
Market price per share	₹ 50	₹ 25
Number of outstanding shares	20 lakhs	10 Lakhs

The merger is expected to generate gains, which have a present value of ₹200 lakhs. The exchange ratio agreed to is 0.5.

What is the true cost of the merger from the point of view of Elrond Limited?

Solution — ICAI

2. Shareholders of Doom Ltd. will get 5 lakh share of Elrond Limited, so they will get:

$$= \frac{5 \text{ lakh}}{20 \text{ lakh} + 5 \text{ lakh}} = 20\% \text{ of shares Elrond Limited}$$

The value of Elrond Ltd. after merger will be:

= ₹ 50 x 20 lakh + ₹ 25 x 10 lakh + ₹ 200 lakh

= ₹ 1000 lakh + ₹ 250 lakh + ₹ 200 lakh = ₹ 1450 lakh

True Cost of Merger will be:

(₹ 1450 x 20%) ₹ 290 lakhs – ₹ 250 lakhs = ₹ 40 lakhs

Practical Q3

3. MK Ltd. is considering acquiring NN Ltd. The following information is available:

Company	Earning after Tax (₹)	No. of Equity Shares	Market Value Per Share (₹)
MK Ltd.	60,00,000	12,00,000	200.00
NN Ltd.	18,00,000	3,00,000	160.00

Exchange of equity shares for acquisition is based on current market value as above. There is no synergy advantage available.

(i) Find the earning per share for company MK Ltd. after merger, and

(ii) Find the exchange ratio so that shareholders of NN Ltd. would not be at a loss.

Solution — ICAI

3. (i) Earning per share of company MK Ltd after merger:-

Exchange ratio 160 : 200 = 4 : 5.

that is 4 shares of MK Ltd. for every 5 shares of NN Ltd.

∴ Total number of shares to be issued = 4/5 × 3,00,000 = 2,40,000 Shares.

∴ Total number of shares of MK Ltd. and NN Ltd.=12,00,000 (MK Ltd.)+ 2,40,000 (NN Ltd.)

= 14,40,000 Shares

Total profit after tax = ₹ 60,00,000 MK Ltd.

= ₹ 18,00,000 NN Ltd.

= ₹ 78,00,000

∴ EPS. (Earning Per Share) of MK Ltd. after merger

₹ 78,00,000/14,40,000 = ₹ 5.42 per share

(ii) To find the exchange ratio so that shareholders of NN Ltd. would not be at a Loss:

Present earning per share for company MK Ltd.

= ₹ 60,00,000/12,00,000 = ₹ 5.00

Present earning per share for company NN Ltd.

= ₹ 18,00,000/3,00,000 = ₹ 6.00

∴ Exchange ratio should be 6 shares of MK Ltd. for every 5 shares of NN Ltd.

∴ Shares to be issued to NN Ltd. = 3,00,000 × 6/5 = 3,60,000 shares

Now, total No. of shares of MK Ltd. and NN Ltd. =12,00,000 (MK Ltd.) + 3,60,000 (NN Ltd.)

= 15,60,000 shares

∴ EPS after merger = ₹ 78,00,000/15,60,000 = ₹ 5.00 per share

Total earnings available to shareholders of NN Ltd. after merger = 3,60,000 shares × ₹ 5.00 = ₹ 18,00,000.

This is equal to earnings prior merger for NN Ltd.

∴ Exchange ratio on the basis of earnings per share is recommended.

Practical Q4

4. ABC Ltd. is intending to acquire XYZ Ltd. by merger and the following information is available in respect of the companies:

	ABC Ltd.	XYZ Ltd.
Number of equity shares	10,00,000	6,00,000
Earnings after tax (₹)	50,00,000	18,00,000
Market value per share (₹)	42	28

Required:

(i) What is the present EPS of both the companies?

(ii) If the proposed merger takes place, what would be the new earning per share for ABC Ltd.? Assume that the merger takes place by exchange of equity shares and the exchange ratio is based on the current market price.

(iii) What should be exchange ratio, if XYZ Ltd. wants to ensure the earnings to members are same as before the merger takes place?

Solution — ICAI

4. (i) Earnings per share = Earnings after tax /No. of equity shares

ABC Ltd. = ₹ 50,00,000/10,00,000 = ₹ 5

XYZ Ltd. = ₹ 18,00,000 / 6,00,000 = ₹ 3

(ii) Number of Shares XYZ Limited's shareholders will get in ABC Ltd. based on market value per share = ₹ 28/ 42 × 6,00,000 = 4,00,000 shares

Total number of equity shares of ABC Ltd. after merger = 10,00,000 + 4,00,000 = 14,00,000 shares

Earnings per share after merger = ₹ 50,00,000 + 18,00,000/14,00,000 = ₹ 4.86

(iii) Calculation of exchange ratio to ensure shareholders of XYZ Ltd. to earn the same as was before merger:

Shares to be exchanged based on EPS = (₹ 3/₹ 5) × 6,00,000 = 3,60,000 shares

EPS after merger = (₹ 50,00,000 + 18,00,000)/13,60,000 = ₹ 5

Total earnings in ABC Ltd. available to shareholders of XYZ Ltd. = 3,60,000 × ₹ 5 = ₹ 18,00,000.

Thus, to ensure that Earning to members are same as before, the ratio of exchange should be 0.6 share for 1 share.

Practical QS

5. The CEO of a company thinks that shareholders always look for EPS. Therefore, he considers maximization of EPS as his company's objective. His company's current Net Profits are ₹ 80.00 lakhs and P/E multiple is 10.5. He wants to buy another firm which has current income of ₹ 15.75 lakhs & P/E multiple of 10.

What is the maximum exchange ratio which the CEO should offer so that he could keep EPS at the current level, given that the current market price of both the acquirer and the target company are ₹ 42 and ₹ 105 respectively?

If the CEO borrows funds at 15% and buys out Target Company by paying cash, how much cash should he offer to maintain his EPS? Assume tax rate of 30%.

Solution — ICAI

5. (i)

	Acquirer Company	Target Company
Net Profit	₹ 80 lakhs	₹ 15.75 lakhs
PE Multiple	10.50	10.00
Market Capitalization	₹ 840 lakhs	₹ 157.50 lakhs
Market Price	₹ 42	₹ 105
No. of Shares	20 lakhs	1.50 lakhs
EPS	₹ 4	₹ 10.50

Maximum Exchange Ratio 4 : 10.50 or 1 : 2.625

Thus, for every one share of Target Company 2.625 shares of Acquirer Company.

(ii) Let x lakhs be the amount paid by Acquirer company to Target Company. Then to maintain same EPS i.e. ₹ 4 the number of shares to be issued will be:

$$\frac{(80 \text{ lakhs} + 15.75 \text{ lakhs}) - 0.70 \times 15\% \times x}{20 \text{ lakhs}} = 4$$

$$\frac{95.75 - 0.105x}{20} = 4$$

x = ₹ 150 lakhs

Thus, ₹ 150 lakhs shall be offered in cash to Target Company to maintain same EPS.

Practical Q6

6. A Ltd. wants to acquire T Ltd. and has offered a swap ratio of 1:2 (0.5 shares for every one share of T Ltd.). Following information is provided:

	A Ltd.	T. Ltd.
Profit after tax	₹18,00,000	₹3,60,000
Equity shares outstanding (Nos.)	6,00,000	1,80,000
EPS	₹3	₹2
PE Ratio	10 times	7 times
Market price per share	₹30	₹14

Required:

(i) The number of equity shares to be issued by A Ltd. for acquisition of T Ltd.

(ii) What is the EPS of A Ltd. after the acquisition?

(iii) Determine the equivalent earnings per share of T Ltd.

(iv) What is the expected market price per share of A Ltd. after the acquisition, assuming its PE multiple remains unchanged?

(v) Determine the market value of the merged firm.

Solution — ICAI

6. (i) The number of shares to be issued by A Ltd.:

The Exchange ratio is 0.5

So, new Shares = 1,80,000 x 0.5 = 90,000 shares.

(ii) **EPS of A Ltd. After a acquisition:**

Total Earnings	(₹ 18,00,000 + ₹ 3,60,000)	₹ 21,60,000
No. of Shares	(6,00,000 + 90,000)	6,90,000
EPS	(₹ 21,60,000)/6,90,000)	₹ 3.13

(iii) **Equivalent EPS of T Ltd.:**

No. of new Shares	0.5
EPS	₹ 3.13
Equivalent EPS (₹ 3.13 x 0.5)	₹ 1.57

(iv) **New Market Price of A Ltd. (P/E remaining unchanged):**

Present P/E Ratio of A Ltd.	10 times
Expected EPS after merger	₹ 3.13
Expected Market Price (₹3.13 x 10)	₹ 31.30

(v) **Market Value of merged firm:**

Total number of Shares	6,90,000
Expected Market Price	₹ 31.30
Total value (6,90,000 x 31.30)	₹ 2,15,97,000

Practical Q7

7. The following information is provided related to the acquiring Firm Mark Limited and the target Firm Mask Limited:

	Firm Mark Limited	Firm Mask Limited
Earning after tax (₹)	2,000 lakhs	400 lakhs
Number of shares outstanding	200 lakhs	100 lakhs
P/E ratio (times)	10	5

Required:

(i) What is the Swap Ratio based on current market prices?

(ii) What is the EPS of Mark Limited after acquisition?

(iii) What is the expected market price per share of Mark Limited after acquisition, assuming P/E ratio of Mark Limited remains unchanged?

(iv) Determine the market value of the merged firm.

(v) Calculate gain/loss for shareholders of the two independent companies after acquisition.

Solution — ICAI

7.

Particulars	Mark Ltd.	Mask Ltd.
EPS	₹ 2,000 Lakhs/ 200 lakhs	₹ 400 lakhs / 100 lakhs

	= ₹ 10	₹ 4
Market Price	₹ 10 × 10 = ₹ 100	₹ 4 × 5 = ₹ 20

(i) The Swap ratio based on current market price is

₹ 20 / ₹ 100 = 0.2 or 1 share of Mark Ltd. for 5 shares of Mask Ltd.

No. of shares to be issued = 100 lakh × 0.2 = 20 lakhs.

(ii) EPS after merger = $\frac{₹ \text{ 2,000 lakhs} + ₹ \text{ 400 lakhs}}{\text{200 lakhs} + \text{20 lakhs}}$ = ₹ 10.91

(iii) Expected market price after merger assuming P/E 10 times.

= ₹ 10.91 × 10 = ₹ 109.10

(iv) Market value of merged firm

= ₹ 109.10 market price × 220 lakhs shares = 240.02 crores

(v) Gain from the merger

Post merger market value of the merged firm	₹ 240.02 crores
Less: Pre-merger market value	
Mark Ltd. 200 Lakhs × ₹ 100 = 200 crores	
Mask Ltd. 100 Lakhs × ₹ 20 = 20 crores	₹ 220.00 crores
Gain from merger	₹ 20.02 crores

Appropriation of gains from the merger among shareholders:

	Mark Ltd.	Mask Ltd.
Post merger value	218.20 crores	21.82 crores
Less: Pre-merger market value	200.00 crores	20.00 crores
Gain to Shareholders	18.20 crores	1.82 crores

Practical Q8

8. XYZ Ltd. wants to purchase ABC Ltd. by exchanging 0.7 of its shares for each share of ABC Ltd. Relevant financial data are as follows:

Equity shares outstanding	10,00,000	4,00,000
EPS (₹)	40	28
Market price per share (₹)	250	160

(i) Illustrate the impact of merger on EPS of both the companies.

(ii) The management of ABC Ltd. has quoted a share exchange ratio of 1:1 for the merger. Assuming that P/E ratio of XYZ Ltd. will remain unchanged after the merger, what will be the gain from merger for ABC Ltd.?

(iii) What will be the gain/loss to shareholders of XYZ Ltd.?

(iv) Determine the maximum exchange ratio acceptable to shareholders of XYZ Ltd.

Solution — ICAI

8. Working Notes

(a)

	XYZ Ltd.	ABC Ltd.
Equity shares outstanding (Nos.)	10,00,000	4,00,000
EPS	₹ 40	₹ 28
Profit	₹ 400,00,000	₹ 112,00,000
PE Ratio	6.25	5.71
Market price per share	₹ 250	₹ 160

(b) EPS after merger

No. of shares to be issued (4,00,000 x 0.70)	2,80,000
Exiting Equity shares outstanding	10,00,000
Equity shares outstanding after merger	12,80,000
Total Profit (₹ 400,00,000 + ₹ 112,00,000)	₹ 512,00,000
EPS	₹ 40

(i) Impact of merger on EPS of both the companies

	XYZ Ltd.	ABC Ltd.
EPS after Merger	₹ 40	₹ 28
EPS before Merger	₹ 40	₹ 28*
	Nil	Nil

* ₹ 40 x 0.70

(ii) Gain from the Merger if exchange ratio is 1: 1

No. of shares to be issued	4,00,000
Exiting Equity shares outstanding	10,00,000
Equity shares outstanding after merger	14,00,000
Total Profit (₹ 400,00,000 + ₹ 112,00,000)	₹ 512,00,000
EPS	₹ 36.57
Market Price of Share (₹ 36.57 x 6.25)	₹ 228.56
Market Price of Share before Merger	₹ 160.00
Impact (Increase/ Gain)	₹ 68.56

(iii) Gain/ loss from the Merger to the shareholders of XYZ Ltd.

Market Price of Share	₹ 228.56
Market Price of Share before Merger	₹ 250.00
Loss from the merger (per share)	₹ 21.44

(iv) Maximum Exchange Ratio acceptable to XYZ Ltd. shareholders

	₹ Lakhs
Market Value of Merged Entity (₹ 228.57 x 1400000)	3199.98
Less: Value acceptable to shareholders of XYZ Ltd.	2500.00
Value of merged entity available to shareholders of ABC Ltd.	699.98
Market Price Per Share	250
No. of shares to be issued to the shareholders of ABC Ltd. (lakhs)	2.80

Thus maximum ratio of issue shall be 2.80 : 4.00 or 0.70 share of XYZ Ltd. for one share of ABC Ltd.

Alternatively, it can also be computed as follows:

Earning after Merger (40 x 1000000 + 28 x 400000)	₹ 512 lakhs
PE Ratio of XYZ Ltd.	6.25
Market Value of Firm after Merger (512 x 6.25)	₹ 3200 lakhs
Existing Value of Shareholders of XYZ Ltd.	₹ 2500 lakhs
Value of Merged entity available to Shareholders of ABC Ltd.	₹ 700 lakhs
Market Price per Share	₹ 250
Total No. of shares to be issued	2.8 lakh

Thus, maximum acceptable ratio shall be 2.80:4.00 i.e. 0.70 share of XYZ Ltd. for one share of ABC Ltd.

Practical Q9

9. XYZ Ltd., is considering merger with ABC Ltd. XYZ Ltd.'s shares are currently traded at ₹ 20. It has 2,50,000 shares outstanding and its earnings after taxes (EAT) amount to ₹ 5,00,000. ABC Ltd., has 1,25,000 shares outstanding; its current market price is ₹ 10 and its EAT are ₹ 1,25,000. The merger will be effected by means of a stock swap (exchange). ABC Ltd., has agreed to a plan under which XYZ Ltd., will offer the current market value of ABC Ltd.'s shares:

(i) What is the pre-merger earnings per share (EPS) and P/E ratios of both the companies?

(ii) If ABC Ltd.'s P/E ratio is 6.4, what is its current market price? What is the exchange ratio? What will XYZ Ltd.'s post-merger EPS be?

(iii) What should be the exchange ratio; if XYZ Ltd.'s pre-merger and post-merger EPS are to be the same?

Solution — ICAI

9. (i) Pre-merger EPS and P/E ratios of XYZ Ltd. and ABC Ltd.

Particulars	XYZ Ltd.	ABC Ltd.
Earnings after taxes	5,00,000	1,25,000
Number of shares outstanding	2,50,000	1,25,000
EPS	2	1
Market Price per share	20	10
P/E Ratio (times)	10	10

(ii) Current Market Price of ABC Ltd. if P/E ratio is 6.4 = ₹ 1 × 6.4 = ₹ 6.40

Exchange ratio = $\frac{₹\ 20}{₹\ 6.40} = 3.125$ or $\frac{₹\ 6.40}{₹\ 20} = 0.32$

Post-merger EPS of XYZ Ltd.

$$= \frac{₹\ 5,00,000 + ₹\ 1,25,000}{2,50,000 + (1,25,000/3.125)}$$

$$= \frac{₹\ 6,25,000}{2,90,000} = 2.16$$

(iii) Desired Exchange Ratio

Total number of shares in post-merged company

$$= \frac{\text{Post-merger earnings}}{\text{Pre-merger EPS of XYZ Ltd}} = \frac{₹\ 6,25,000}{2} = 3,12,500$$

Number of shares required to be issued

= 3,12,500 – 2,50,000 = 62,500

Therefore, the exchange ratio is

62,500 : 1,25,000

$$= \frac{62,500}{1,25,000} = 0.50$$

Practical Q10

10. Following information is provided relating to the acquiring company Mani Ltd. and the target company Ratnam Ltd:

	Mani Ltd.	Ratnam Ltd.
Earnings after tax (₹ lakhs)	2,000	4,000
No. of shares outstanding (lakhs)	200	1,000

P/E ratio (No. of times)	10	5

Required:

(i) What is the swap ratio based on current market prices?

(ii) What is the EPS of Mani Ltd. after the acquisition?

(iii) What is the expected market price per share of Mani Ltd. after the acquisition, assuming its P/E ratio is adversely affected by 10%?

(iv) Determine the market value of the merged Co.

(v) Calculate gain/loss for the shareholders of the two independent entities, due to the merger.

Solution — ICAI

10. (i) SWAP ratio based on current market prices:

EPS before acquisition:	
Mani Ltd. : ₹2,000 lakhs / 200 lakhs:	₹10
Ratnam Ltd.: ₹4,000 lakhs / 1,000 lakhs:	₹ 4
Market price before acquisition:	
Mani Ltd.: ₹10 × 10	₹100
Ratnam Ltd.: ₹4 × 5	₹ 20
SWAP ratio: 20/100 or 1/5 i.e.	0.20

(ii) EPS after acquisition:

$$\frac{₹(2,000+4,000)\ \text{Lakhs}}{(200+200)\ \text{Lakhs}} = ₹15.00$$

(iii) Market Price after acquisition:

EPS after acquisition :	₹15.00
P/E ratio after acquisition 10 × 0.9	9
Market price of share (₹ 15 X 9)	₹135.00

(iv) Market value of the merged Co.:

₹135 × 400 lakhs shares	₹ 540.00 Crores
	or ₹ 54,000 Lakhs

(v) Gain/loss per share:

	₹ Crore	
	Mani Ltd.	*Ratnam Ltd.*
Total value before Acquisition	200	200
Value after acquisition	270	270
Gain (Total)	70	70
No. of shares (pre-merger) (lakhs)	200	1,000
Gain per share (₹)	35	7

Practical Q11

11. You have been provided the following Financial data of two companies:

	Krishna Ltd.	Rama Ltd.
Earnings after taxes	₹ 7,00,000	₹ 10,00,000
No. of Equity shares (outstanding)	2,00,000	4,00,000
EPS	3.5	2.5
P/E ratio	10 times	14 times
Market price per share	₹ 35	₹ 35

Company Rama Ltd. is acquiring the company Krishna Ltd., exchanging its shares on a one-to-one basis for company Krishna Ltd. The exchange ratio is based on the market prices of the shares of the two companies.

Required:

(i) What will be the EPS subsequent to merger?

(ii) What is the change in EPS for the shareholders of companies Rama Ltd. and Krishna Ltd.?

(iii) Determine the market value of the post-merger firm. PE ratio is likely to remain the same.

(iv) Ascertain the profits accruing to shareholders of both the companies.

Solution — ICAI

11.

(i)	Exchange Ratio	1:1
	New Shares to be issued	2,00,000
	Total shares of Rama Ltd. (4,00,000+2,00,000)	6,00,000
	Total earnings (₹ 10,00,000 + ₹ 7,00,000)	₹ 17,00,000
	New EPS (₹ 17,00,000/6,00,000)	₹ 2.83
(ii)	Existing EPS of Rama Ltd.	₹ 2.50
	Increase in EPS of Rama Ltd (₹ 2.83 – ₹ 2.50)	₹ 0.33
	Existing EPS of Krishna Ltd.	₹ 3.50
	Decrease in EPS of Krishna Ltd. (₹ 3.50 – ₹ 2.83)	₹ 0.67
(iii)	P/E ratio of new firm (expected to remain same)	14 times
	New market price (14 × ₹ 2.83)	₹ 39.62
	Total No. of Shares	6,00,000
	Total market Capitalization (6,00,000 × ₹ 39.62)	₹ 2,37,72,000
	Existing market capitalization (₹ 70,00,000 + ₹ 1,40,00,000)	₹ 2,10,00,000
	Total gain	₹ 27,72,000

(iv)

	Rama Ltd.	Krishna Ltd	Total
No. of shares after merger	4,00,000	2,00,000	6,00,000
Market price	₹ 39.62	₹ 39.62	₹ 39.62
Total Mkt. Values	₹ 1,58,48,000	₹ 79,24,000	₹ 2,37,72,000
Existing Mkt. values	₹ 1,40,00,000	₹ 70,00,000	₹ 2,10,00,000
Gain to share holders	₹ 18,48,000	₹ 9,24,000	₹ 27,72,000

or ₹ 27,72,000 ÷ 3 = ₹ 9,24,000 to Krishna Ltd. and ₹ 18,48,000 to Rama Ltd. (in 2: 1 ratio)

Practical Q12

12. M Co. Ltd. is studying the possible acquisition of N Co. Ltd., by way of merger. The following data are available in respect of the companies:

Particulars	M Co. Ltd.	N Co. Ltd.
Earnings after tax (₹)	80,00,000	24,00,000
No. of equity shares	16,00,000	4,00,000
Market value per share (₹)	200	160

(i) If the merger goes through by exchange of equity and the exchange ratio is based on the current market price, what is the new earning per share for M Co. Ltd.?

(ii) N Co. Ltd. wants to be sure that the earnings available to its shareholders will not be diminished by the merger. What should be the exchange ratio in that case?

Solution — ICAI

12. **(i)** Calculation of new EPS of M Co. Ltd.

No. of equity shares to be issued by M Co. Ltd. to N Co. Ltd.

= 4,00,000 shares × ₹ 160/₹ 200 = 3,20,000 shares

Total no. of shares in M Co. Ltd. after acquisition of N Co. Ltd.

= 16,00,000 + 3,20,000 = 19,20,000

Total earnings after tax [after acquisition]

= 80,00,000 + 24,00,000 = 1,04,00,000

$$\text{EPS} = \frac{₹\ 1{,}04{,}00{,}000}{19{,}20{,}000 \text{ equity shares}} = ₹\ 5.42$$

(ii) Calculation of exchange ratio which would not diminish the EPS of N Co. Ltd. after its merger with M Co. Ltd.

Current EPS:

$$\text{M Co. Ltd.} = \frac{₹\ 80{,}00{,}000}{16{,}00{,}000 \text{ equity shares}} = ₹\ 5$$

$$\text{N Co. Ltd.} = \frac{₹\ 24{,}00{,}000}{4{,}00{,}000 \text{ equity shares}} = ₹\ 6$$

Exchange ratio = 6/5 = 1.20

No. of new shares to be issued by M Co. Ltd. to N Co. Ltd.

= 4,00,000 × 1.20 = 4,80,000 shares

Total number of shares of M Co. Ltd. after acquisition

= 16,00,000 + 4,80,000 = 20,80,000 shares

$$\text{EPS [after merger]} = \frac{₹\ 1{,}04{,}00{,}000}{20{,}80{,}000 \text{ shares}} = ₹\ 5$$

Total earnings in M Co. Ltd. available to new shareholders of N Co. Ltd.

= 4,80,000 × ₹ 5 = ₹ 24,00,000

Recommendation: The exchange ratio (6 for 5) based on market shares is beneficial to shareholders of 'N' Co. Ltd.

Practical Q13

13. Longitude Limited is in the process of acquiring Latitude Limited on a share exchange basis. Following relevant data are available:

		Longitude Limited	Latitude Limited
Profit after Tax (PAT)	₹ in Lakhs	120	80
Number of Shares	Lakhs	15	16
Earning per Share (EPS)	₹	8	5
Price Earnings Ratio (P/E Ratio)		15	10
(Ignore Synergy)			

You are required to determine:

(i) Pre-merger Market Value per Share, and

(ii) The maximum exchange ratio Longitude Limited can offer without the dilution of

(1) EPS and

(2) Market Value per Share

Calculate Ratio/s up to four decimal points and amounts and number of shares up to two decimal points.

Solution — ICAI

13. (i) Pre Merger Market Value of Per Share

P/E Ratio X EPS

Longitude Ltd. ₹ 8 X 15 = ₹ 120.00

Latitude Ltd. ₹ 5 X 10 = ₹ 50.00

(ii) (1) Maximum exchange ratio without dilution of EPS

Pre Merger PAT of Longitude Ltd.	₹ 120 Lakhs
Pre Merger PAT of Latitude Ltd.	₹ 80 Lakhs
Combined PAT	₹ 200 Lakhs
Longitude Ltd. 's EPS	₹ 8
Maximum number of shares of Longitude after merger (₹ 200 lakhs/₹ 8)	25 Lakhs
Existing number of shares	15 Lakhs
Maximum number of shares to be exchanged	10 Lakhs

Maximum share exchange ratio 10:16 or 5:8

(2) Maximum exchange ratio without dilution of Market Price Per Share

Pre Merger Market Capitalization of Longitude Ltd. (₹ 120 × 15 Lakhs)	₹ 1800 Lakhs
Pre Merger Market Capitalization of Latitude Ltd. (₹ 50 × 16 Lakhs)	₹ 800 Lakhs
Combined Market Capitalization	₹ 2600 Lakhs
Current Market Price of share of Longitude Ltd.	₹ 120
Maximum number of shares to be exchanged of Longitude (surviving company) (₹ 2600 Lakhs/₹ 120)	21.67 Lakhs
Current Number of Shares of Longitude Ltd.	15.00 Lakhs
Maximum number of shares to be exchanged (Lakhs)	6.67 Lakhs

Maximum share exchange ratio 6.67:16 or 0.4169:1

Practical Q14

14. P Ltd. is considering take-over of R Ltd. by the exchange of four new shares in P Ltd. for every five shares in R Ltd. The relevant financial details of the two companies prior to merger announcement are as follows:

	P Ltd	R Ltd
Profit before Tax (₹ Crore)	15	13.50
No. of Shares (Crore)	25	15
P/E Ratio	12	9

Corporate Tax Rate 30%

You are required to determine:

(i) Market value of both the company.

(ii) Value of original shareholders.

(iii) Price per share after merger.

(iv) Effect on share price of both the company if the Directors of P Ltd. expect their own pre-merger P/E ratio to be applied to the combined earnings.

Solution — ICAI

14.

	P Ltd.	R Ltd.
Profit before Tax (₹ in crore)	15	13.50
Tax 30% (₹ in crore)	4.50	4.05
Profit after Tax (₹ in crore)	10.50	9.45
Earning per Share (₹)	$\frac{10.50}{25}$ = ₹ 0.42	$\frac{9.45}{15}$ = ₹ 0.63
Price of Share before Merger (EPS x P/E Ratio)	₹ 0.42 x 12 = ₹ 5.04	₹0.63 x 9 = ₹ 5.67

(i) ∴ Market Value of company

P Ltd. = ₹ 5.04 x 25 Crore = ₹ 126 crore

R Ltd. = ₹ 5.67 x 15 Crore = ₹ 85.05 crore

Combined = ₹ 126 + ₹ 85.05 = ₹ 211.05 Crores

After Merger

	P Ltd.	R Ltd.
No. of Shares	25 crores	$15x\frac{4}{5}$ = 12 crores
Combined	37 crores	
% of Combined Equity Owned	$\frac{25}{37}x100 = 67.57\%$	$\frac{12}{37}x100 = 32.43\%$

(ii) ∴ Value of Original Shareholders

P Ltd.	R Ltd.
₹ 211.05 crore x 67.57%	₹ 211.05 crore x 32.43%
= ₹ 142.61	= ₹ 68.44

Alternatively, it can also be computed as follows:

Combined Value of Entity	211.05 crore
No. of shares after Merger	37 crore
Value of Per Share	₹ 5.70405
Value of P Ltd. Shareholders (25 crores x ₹ 5.70405)	₹ 142.60 crore
Value of R Ltd. Shareholders (12 crores x ₹ 5.70405)	₹ 68.45 crore

(iii) ∴ Price per Share after Merger

$$EPS = \frac{₹19.95\,crore}{37\,crore} = ₹\ 0.539 \text{ per share}$$

P/E Ratio = 12

Market Value Per Share = ₹ 0.539 X 12 = ₹ 6.47

Total Market Value = ₹ 6.47 x 37 crore = ₹ 239.39 crore

$$\text{Price of Share} = \frac{\text{Market Value}}{\text{Number of Shares}} = \frac{239.39\,crore}{37\,crore} = ₹\ 6.47$$

(iv) Effect on Share Price

P Ltd.

Gain/loss (-) per share = ₹ 6.47 – ₹ 5.04 = ₹ 1.43

i.e. $\frac{6.47-5.04}{5.04} \times 100 = 0.284$ or 28.4%

∴ Share price would rise by 28.4%

R Ltd.

$6.47 \times \frac{4}{5} = ₹\ 5.18$

Gain/loss (-) per share = ₹ 5.18 – ₹ 5.67 = (-₹ 0.49)

i.e. $\frac{5.18-5.67}{5.67} \times 100$ (-) 0.0864 or (-) 8.64%

∴ Share Price would decrease by 8.64%.

Practical QIS

15. Simple Ltd. and Dimple Ltd. are planning to merge. The total value of the companies are dependent on the fluctuating business conditions. The following information is given for the total value (debt + equity) structure of each of the two companies.

Business Condition	Probability	Simple Ltd. ₹ Lacs	Dimple Ltd. ₹ Lacs
High Growth	0.20	820	1050
Medium Growth	0.60	550	825
Slow Growth	0.20	410	590

The current debt of Dimple Ltd. is ₹ 65 lacs and of Simple Ltd. is ₹ 460 lacs.

Calculate the expected value of debt and equity separately for the merged entity.

Solution — ICAI

15. Compute Value of Equity

Simple Ltd.

₹ in Lacs

	High Growth	Medium Growth	Slow Growth
Debt + Equity	820	550	410
Less: Debt	460	460	460
Equity	360	90	-50

Since the Company has limited liability the value of equity cannot be negative therefore the value of equity under slow growth will be taken as zero because of insolvency risk and the value of debt is taken at 410 lacs. The expected value of debt and equity can then be calculated as:

Simple Ltd.

₹ in Lacs

	High Growth		Medium Growth		Slow Growth		Expected Value
	Prob.	Value	Prob.	Value	Prob.	Value	
Debt	0.20	460	0.60	460	0.20	410	450
Equity	0.20	360	0.60	90	0.20	0	126
		820		550		410	576

Dimple Ltd.

₹ in Lacs

	High Growth		Medium Growth		Slow Growth		Expected Value
	Prob.	Value	Prob.	Value	Prob.	Value	
Equity	0.20	985	0.60	760	0.20	525	758
Debt	0.20	65	0.60	65	0.20	65	65
		1050		825		590	823

Expected Values

₹ in Lacs

Equity		Debt	
Simple Ltd.	126	Simple Ltd.	450
Dimple Ltd.	758	Dimple Ltd.	65
	884		515

Practical Q16

16. Yes Ltd. wants to acquire No Ltd. and the cash flows of Yes Ltd. and the merged entity are given below:

(₹ In lakhs)

Year	1	2	3	4	5
Yes Ltd.	175	200	320	340	350
Merged Entity	400	450	525	590	620

Earnings would have witnessed 5% constant growth rate without merger and 6% with merger on account of economies of operations after 5 years in each case. The cost of capital is 15%.

The number of shares outstanding in both the companies before the merger is the same and the companies agree to an exchange ratio of 0.5 shares of Yes Ltd. for each share of No Ltd.

PV factor at 15% for years 1-5 are 0.870, 0.756; 0.658, 0.572, 0.497 respectively.

You are required to:

(i) Compute the Value of Yes Ltd. before and after merger.

(ii) Value of Acquisition and

(iii) Gain to shareholders of Yes Ltd.

Solution — ICAI

16. (i) **Working Notes:**

Present Value of Cash Flows (CF) upto 5 years

Year End	CF of Yes Ltd. (₹ lakhs)	PVF @15%	PV of CF (₹ lakhs)	CF of Merged Entity (₹ lakhs)	PV of CF of Merged Entity (₹ lakhs)
1	175	0.870	152.25	400	348.00
2	200	0.756	151.20	450	340.20
3	320	0.658	210.56	525	345.45

4	340	0.572	194.48	590	337.48
5	350	0.497	173.95	620	308.14
			882.44		1679.27

PV of Cash Flows of Yes Ltd. after the forecast period

$$TV_5 = \frac{CF_5(1+g)}{K_e - g} = \frac{350(1+0.05)}{0.15-0.05} = \frac{367.50}{0.10} = ₹\ 3675 \text{ lakhs}$$

PV of TV_5 = ₹3675 lakhs x 0.497 = ₹ 1826.475 lakhs

PV of Cash Flows of Merged Entity after the forecast period

$$TV_5 = \frac{CF_5(1+g)}{K_e - g} = \frac{620(1+0.06)}{0.15-0.06} = \frac{657.20}{0.09} = ₹\ 7302.22 \text{ lakhs}$$

PV of TV_5 = ₹ 7302.22 lakhs x 0.497 = ₹ 3629.20 lakhs

Value of Yes Ltd.

	Before merger (₹ lakhs)	After merger (₹ lakhs)
PV of CF (1-5 years)	882.440	1679.27
Add: PV of TV_5	1826.475	3629.20
	2708.915	5308.47

(ii) Value of Acquisition

= Value of Merged Entity – Value of Yes Ltd.

= ₹ 5308.47 lakhs – ₹ 2708.915 lakhs = ₹ 2599.555 lakhs

(iii) Gain to Shareholders of Yes Ltd.

Share of Yes Ltd. in merged entity = ₹ 5308.47 lakhs x $\frac{1}{1.5}$ = ₹ 3538.98 lakhs

Gain to shareholder = Share of Yes Ltd. in merged entity – Value of Yes Ltd. before merger

= ₹ 3538.98 lakhs - ₹ 2708.915 = ₹ 830.065 lakhs

Practical Q17

17. The following information is provided relating to the acquiring company Efficient Ltd. and the target Company Healthy Ltd.

	Efficient Ltd.	Healthy Ltd.
No. of shares (F.V. ₹ 10 each)	10.00 lakhs	7.5 lakhs
Market capitalization	500.00 lakhs	750.00 lakhs
P/E ratio (times)	10.00	5.00
Reserves and Surplus	300.00 lakhs	165.00 lakhs
Promoter's Holding (No. of shares)	4.75 lakhs	5.00 lakhs

Board of Directors of both the Companies have decided to give a fair deal to the shareholders and accordingly for swap ratio the weights are decided as 40%, 25% and 35% respectively for Earning, Book Value and Market Price of share of each company:

(i) Calculate the swap ratio and also calculate Promoter's holding % after acquisition.

(ii) What is the EPS of Efficient Ltd. after acquisition of Healthy Ltd.?

(iii) What is the expected market price per share and market capitalization of Efficient Ltd. after acquisition, assuming P/E ratio of Firm Efficient Ltd. remains unchanged.

(iv) Calculate free float market capitalization of the merged firm.

Solution — ICAI

17. **Swap Ratio**

	Efficient Ltd.	Healthy Ltd.
Market capitalization	500 lakhs	750 lakhs
No. of shares	10 lakhs	7.5 lakhs

Market Price per share	₹ 50	₹ 100
P/E ratio	10	5
EPS	₹ 5	₹ 20
Profit	₹ 50 lakh	₹ 150 lakh
Share capital	₹ 100 lakh	₹ 75 lakh
Reserves and surplus	₹ 300 lakh	₹ 165 lakh
Total	₹ 400 lakh	₹ 240 lakh
Book Value per share	₹ 40	₹ 32

(i) Calculation of Swap Ratio

EPS	1 : 4 i.e.	4.0 × 40%	1.6
Book value	1 : 0.8 i.e.	0.8 × 25%	0.2
Market price	1 : 2 i.e.	2.0 × 35%	0.7
		Total	2.5

Swap ratio is for every one share of Healthy Ltd., to issue 2.5 shares of Efficient Ltd.
Hence, total no. of shares to be issued 7.5 lakh × 2.5 = 18.75 lakh shares.

Promoter's holding = 4.75 lakh shares + (5 × 2.5 = 12.5 lakh shares) = 17.25 lakh i.e.
Promoter's holding % is (17.25 lakh/28.75 lakh) × 100 = 60%.

Calculation of EPS, Market price, Market capitalization and free float market capitalization.

(ii) Total No. of shares 10 lakh + 18.75 lakh = 28.75 lakh

Total capital 100 lakh + 187.5 lakh = ₹ 287.5 lakh

$$\text{EPS } \frac{\text{Total profit}}{\text{No. of shares}} = \frac{50\,\text{lakh} + 150\,\text{lakh}}{28.75\,\text{lakh}} = \frac{200}{28.75} = ₹\ 6.956$$

(iii) Expected market price EPS 6.956 × P/E 10 = ₹ 69.56

Market capitalization = ₹ 69.56 per share × 28.75 lakh shares

= ₹ 1,999.85 lakh

(iv) Free float of market capitalization = ₹ 69.56 per share × (28.75 lakh × 40%)

= ₹ 799.94 lakh

Practical Q18

18. Abhiman Ltd. is a subsidiary of Janam Ltd. and is acquiring Swabhiman Ltd. which is also a subsidiary of Janam Ltd. The following information is given :

	Abhiman Ltd.	Swabhiman Ltd.
% Shareholding of promoter	50%	60%
Share capital	₹ 200 lacs	100 lacs
Free Reserves and surplus	₹ 900 lacs	600 lacs
Paid up value per share	₹ 100	10
Free float market capitalization	₹ 500 lacs	156 lacs
P/E Ratio (times)	10	4

Janam Ltd., is interested in doing justice to both companies. The following parameters have been assigned by the Board of Janam Ltd., for determining the swap ratio:

Book value	25%
Earning per share	50%
Market price	25%

You are required to compute

(i) The swap ratio.

(ii) The Book Value, Earning Per Share and Expected Market Price of Swabhiman Ltd., (assuming P/E Ratio of Abhiman remains the same and all assets and liabilities of Swabhiman Ltd. are taken over at book value.)

Solution — ICAI

18. SWAP RATIO

	Abhiman Ltd. (₹)	Swabhiman Ltd. (₹)
Share capital	200 lacs	100 lacs
Free reserves & surplus	900 lacs	600 lacs
Total	1100 lacs	700 lacs
No. of shares	2 lacs	10 lacs
Book value for share	₹ 550	₹ 70
Promoters Holding	50%	60%
Non promoters holding	50%	40%
Free float market capitalization (Public)	500 lacs	₹ 156 lacs
Total Market Cap	1000 lacs	390 lacs
No. of shares	2 lacs	10 lacs
Market Price	₹ 500	₹ 39
P/E ratio	10	4
EPS	₹ 50.00	₹ 9.75

Calculation of SWAP Ratio

Book Value	1:0.1273	0.1273 × 25%	0.031825
EPS	1:0.195	0.195 × 50%	0.097500
Market Price	1:0.078	0.078 × 25%	0.019500
Total			0.148825

(i) SWAP Ratio is 0.148825 shares of Abhiman Ltd. for every share of Swabhiman Ltd.

Total No. of shares to be issued = 10 lakh × 0.148825 = 148825 shares

(ii) Book value, EPS & Market Price.

Total No. shares = 200000 +148825=348825

Total capital = ₹200 lakh + ₹148.825 lac = ₹ 348.825 lac

Reserves = ₹ 900 lac + ₹ 551.175 lac = ₹ 1451.175 lac

$$\text{Book value Per Share} = \frac{₹\ 348.825\text{lac} + ₹\ 1451.175\ \text{lac}}{3.48825\ \text{lac}} = ₹\ 516.02$$

or ₹ 516.02 x 0.148825 = ₹ 76.80

$$\text{or} = \frac{\text{Total Capital}}{\text{No. of Shares}} = \frac{1100\text{lac} + 700\ \text{lac}}{348825} = ₹\ 516.02$$

$$\text{EPS} = \frac{\text{Total Profit}}{\text{No. of shares}} = \frac{₹100\ \text{lac} + ₹\ 97.50\ \text{lac}}{3.48825\text{lac}} = ₹\ 56.62$$

or ₹ 56.62 x 0.148825 = ₹ 8.43

Expected market price = ₹ 56.62 × PE Ratio= ₹ 56.62 × 10 = ₹ 566.20

or ₹ 566.20 x 0.148825 = ₹ 84.26

Practical Q19

19. The following information is provided relating to the acquiring company E Ltd., and the target company H Ltd:

Particulars	E Ltd. (₹)	H Ltd. (₹)
Number of shares (Face value ₹ 10 each)	20 Lakhs	15 Lakhs
Market Capitalization	1000 Lakhs	1500 Lakhs
P/E Ratio (times)	10.00	5.00
Reserves and surplus in ₹	600.00 Lakhs	330.00 Lakhs
Promoter's Holding (No. of shares)	9.50 Lakhs	10.00 Lakhs

The Board of Directors of both the companies have decided to give a fair deal to the shareholders. Accordingly, the weights are decided as 40%, 25% and 35% respectively for earnings (EPS), book value and market price of share of each company for swap ratio.

Calculate the following:

(i) Market price per share, earnings per share and Book Value per share;

(ii) Swap ratio;

(iii) Promoter's holding percentage after acquisition;

(iv) EPS of E Ltd. after acquisitions of H Ltd;

(v) Expected market price per share and market capitalization of E Ltd.; after acquisition, assuming P/E ratio of E Ltd. remains unchanged; and

(vi) Free float market capitalization of the merged firm.

Solution — ICAI

19.

(i)

	E Ltd.	H Ltd.
Market capitalisation	1000 lakhs	1500 lakhs
No. of shares	20 lakhs	15 lakhs
Market Price per share	₹ 50	₹ 100
P/E ratio	10	5
EPS	₹ 5	₹ 20
Profit	₹ 100 lakh	₹ 300 lakh
Share capital	₹ 200 lakh	₹ 150 lakh
Reserves and surplus	₹ 600 lakh	₹ 330 lakh
Total	₹ 800 lakh	₹ 480 lakh
Book Value per share	₹ 40	₹ 32

(ii) **Calculation of Swap Ratio**

EPS	1 : 4 i.e.	4.0 × 40%	1.6
Book value	1 : 0.8 i.e.	0.8 × 25%	0.2
Market price	1 : 2 i.e.	2.0 × 35%	0.7
		Total	2.5

Swap ratio is for every one share of H Ltd., to issue 2.5 shares of E Ltd. Hence, total no. of shares to be issued 15 lakh × 2.5 = 37.50 lakh shares

(iii) Promoter's holding = 9.50 lakh shares + (10× 2.5 = 25 lakh shares) = 34.50 lakh i.e. Promoter's holding % is (34.50 lakh/57.50 lakh) × 100 = 60%.

(iv) Calculation of EPS after merger

Total No. of shares 20 lakh + 37.50 lakh = 57.50 lakh

$$\text{EPS } \frac{\text{Total profit}}{\text{No. of shares}} = \frac{100 \text{ lakh} + 300 \text{ lakh}}{57.50 \text{ lakh}} = \frac{400}{57.50} = ₹ 6.956$$

(v) Calculation of Market price and Market capitalization after merger

Expected market price — EPS 6.956 × P/E 10 = ₹ 69.56

Market capitalization — = ₹ 69.56 per share ×57.50 lakh shares

= ₹ 3,999.70 lakh or ₹ 4,000 lakh

(vi) Free float of market capitalization = ₹ 69.56 per share × (57.50 lakh × 40%) = ₹ 1599.88 lakh

Practical Q20

20. The following information relating to the acquiring Company Abhiman Ltd. and the target Company Abhishek Ltd. are available. Both the Companies are promoted by Multinational Company, Trident Ltd. The promoter's holding is 50% and 60% respectively in Abhiman Ltd. and Abhishek Ltd.:

	Abhiman Ltd.	Abhishek Ltd.
Share Capital (₹)	200 lakh	100 lakh
Free Reserve and Surplus (₹)	800 lakh	500 lakh
Paid up Value per share (₹)	100	10
Free float Market Capitalisation (₹)	400 lakh	128 lakh
P/E Ratio (times)	10	4

Trident Ltd. is interested to do justice to the shareholders of both the Companies. For the swap ratio weights are assigned to different parameters by the Board of Directors as follows:

Book Value	25%
EPS (Earning per share)	50%
Market Price	25%

(a) What is the swap ratio based on above weights?

(b) What is the Book Value, EPS and expected Market price of Abhiman Ltd. after acquisition of Abhishek Ltd. (assuming P.E. ratio of Abhiman Ltd. remains unchanged and all assets and liabilities of Abhishek Ltd. are taken over at book value).

(c) Calculate:

(i) Promoter's revised holding in the Abhiman Ltd.

(ii) Free float market capitalization.

(iii) Also calculate No. of Shares, Earning per Share (EPS) and Book Value (B.V.), if after acquisition of Abhishek Ltd., Abhiman Ltd. decided to :

(1) Issue Bonus shares in the ratio of 1 : 2; and

(2) Split the stock (share) as ₹ 5 each fully paid.

Solution — ICAI

20. (a) Swap Ratio

	Abhiman Ltd.	Abhishek Ltd.
Share Capital	200 Lakh	100 Lakh
Free Reserves	800 Lakh	500 Lakh
Total	1000 Lakh	600 Lakh
No. of Shares	2 Lakh	10 Lakh
Book Value per share	₹ 500	₹ 60
Promoter's holding	50%	60%
Non promoter's holding	50%	40%
Free Float Market Cap. i.e. relating to Public's holding	400 Lakh	128 Lakh
Hence Total market Cap.	800 Lakh	320 Lakh
No. of Shares	2 Lakh	10 Lakh
Market Price	₹ 400	₹ 32
P/E Ratio	10	4
EPS	40	8
Profits (₹ 2 X 40 lakh)	₹ 80 lakh	-
(₹ 8 X 10 lakh)	-	₹ 80 lakh

Calculation of Swap Ratio

Book Value	1 : 0.12 i.e.	0.12 x 25%	0.03
EPS	1 : 0.2	0.20 x 50%	0.10
Market Price	1 : 0.08	0.08 x 25%	0.02
		Total	0.15

Swap ratio is for every one share of Abhishek Ltd., to issue 0.15 shares of Abhiman Ltd. Hence total no. of shares to be issued.

10 Lakh x 0.15 = 1.50 lakh shares

(b) Book Value, EPS & Market Price

Total No of Shares 2 Lakh + 1.5 Lakh = 3.5 Lakh

Total Capital ₹ 200 Lakh + ₹ 150 Lakh = ₹ 350 Lakh

Reserves ₹ 800 Lakh + ₹ 450 Lakh = ₹ 1,250 Lakh

Book Value $\frac{₹\ 350\ Lakh + ₹\ 1,250\ Lakh}{3.5\ Lakh}$ = ₹ 457.14 per share

EPS $\frac{Total\ Profit}{No.\ of\ Share} = \frac{₹\ 80\ Lakh + ₹\ 80\ Lakh}{3.5\ Lakh} = \frac{₹\ 160\ Lakh}{3.5}$ = ₹ 45.71

Expected Market PriceEPS (₹ 45.71) x P/E Ratio (10) = ₹ 457.10

(c) (i) Promoter's holding

Promoter's Revised	Abhiman 50% i.e.	1.00 Lakh shares
Holding	Abhishek 60% i.e.	0.90 Lakh shares
	Total	1.90 Lakh shares

Promoter's % = 1.90/3.50 x 100 = 54.29%

(ii) Free Float Market Capitalisation

Free Float Market Capitalisation = (3.5 Lakh – 1.9 Lakh) x ₹ 457.10 = ₹ 731.36 Lakh

(iii) (a) & (b)

Revised Capital	₹ 350 Lakh + ₹ 175 Lakh = ₹ 525 Lakh
No. of shares before Split (F.V ₹ 100)	5.25 Lakh
No. of Shares after Split (F.V. ₹ 5)	5.25 x 20 = 105 Lakh
EPS	160 Lakh / 105 Lakh = 1.523
Book Value	Cap. ₹ 525 Lakh + ₹ 1075 Lakh
	No. of Shares =105 Lakh
	= ₹ 15.238 per share

Practical Q21

21. T Ltd. and E Ltd. are in the same industry. The former is in negotiation for acquisition of the latter. Important information about the two companies as per their latest financial statements is given below:

	T Ltd.	E Ltd.
₹ 10 Equity shares outstanding	12 Lakhs	6 Lakhs
Debt:		
10% Debentures (₹ Lakhs)	580	--
12.5% Institutional Loan (₹ Lakhs)	--	240
Earning before interest, depreciation and tax (EBIDAT) (₹ Lakhs)	400.86	115.71
Market Price/share (₹)	220.00	110.00

T Ltd. plans to offer a price for E Ltd., business as a whole which will be 7 times EBIDAT reduced by outstanding debt, to be discharged by own shares at market price.

E Ltd. is planning to seek one share in T Ltd. for every 2 shares in E Ltd. based on the market price. Tax rate for the two companies may be assumed as 30%.

Calculate and show the following under both alternatives - T Ltd.'s offer and E Ltd.'s plan:

(i) Net consideration payable.

(ii) No. of shares to be issued by T Ltd.

(iii) EPS of T Ltd. after acquisition.

(iv) Expected market price per share of T Ltd. after acquisition.

(v) State briefly the advantages to T Ltd. from the acquisition.

Note: Calculations (except EPS) may be rounded off to 2 decimals in lakhs.

Solution — ICAI

21. As per T Ltd.'s Offer

		₹ in lakhs
(i)	**Net Consideration Payable**	
	7 times EBIDAT, i.e. 7 x ₹ 115.71 lakh	809.97
	Less: Debt	240.00
		569.97
(ii)	**No. of shares to be issued by T Ltd**	
	₹ 569.97 lakh/₹ 220 (rounded off) (Nos.)	2,59,000
(iii)	**EPS of T Ltd after acquisition**	
	Total EBIDT (₹ 400.86 lakh + ₹ 115.71 lakh)	516.57
	Less: Interest (₹ 58 lakh + ₹ 30 lakh)	88.00
		428.57
	Less: 30% Tax	128.57
	Total earnings (NPAT)	300.00
	Total no. of shares outstanding	14.59 lakh
	(12 lakh + 2.59 lakh)	
	EPS (₹ 300 lakh/ 14.59 lakh)	₹ 20.56
(iv)	**Expected Market Price:**	
	Pre-acquisition P/E multiple:	
	EBIDAT (₹ in lakhs)	400.86
	Less: Interest $(580 \times \frac{10}{100})$ (₹ in lakhs)	58.00
		342.86
	Less: 30% Tax (₹ in lakhs)	102.86
	EAT (₹ in lakhs)	240.00
	No. of shares (lakhs)	12.00
	EPS	₹ 20.00
	Hence, PE multiple $\frac{220}{20}$	11
	Expected market price after acquisition (₹ 20.56 x 11)	₹ 226.16

As per E Ltd's Plan

		₹ in lakhs
(i)	**Net consideration payable**	
	6 lakhs shares x ₹ 110	660
(ii)	**No. of shares to be issued by T Ltd**	
	₹ 660 lakhs ÷ ₹ 220	3 lakh
(iii)	**EPS of T Ltd after Acquisition**	
	NPAT (as per earlier calculations)	300.00
	Total no. of shares outstanding (12 lakhs + 3 lakhs)	15 lakh
	Earning Per Share (EPS) ₹ 300 lakh/15 lakh	₹ 20.00
(iv)	**Expected Market Price** (₹ 20 x 11)	₹ 220.00

(v) **Advantages of Acquisition to T Ltd**

Since the two companies are in the same industry, the following advantages could accrue:

— Synergy, cost reduction and operating efficiency.

— Better market share.

— Avoidance of competition

Practical Q22

22. The following information is relating to Fortune India Ltd. having two division, viz. Pharma Division and Fast Moving Consumer Goods Division (FMCG Division). Paid up share capital of Fortune India Ltd. is consisting of 3,000 Lakhs equity shares of Re. 1 each. Fortune India Ltd. decided to de-merge Pharma Division as Fortune Pharma Ltd. w.e.f. 1.4.2009. Details of Fortune India Ltd. as on 31.3.2009 and of Fortune Pharma Ltd. as on 1.4.2009 are given below:

Particulars	Fortune Pharma Ltd. ₹	Fortune India Ltd. ₹
Outside Liabilities		
Secured Loans	400 lakh	3,000 lakh
Unsecured Loans	2,400 lakh	800 lakh
Current Liabilities & Provisions	1,300 lakh	21,200 lakh
Assets		
Fixed Assets	7,740 lakh	20,400 lakh
Investments	7,600 lakh	12,300 lakh
Current Assets	8,800 lakh	30,200 lakh
Loans & Advances	900 lakh	7,300 lakh
Deferred tax/Misc. Expenses	60 lakh	(200) lakh

Board of Directors of the Company have decided to issue necessary equity shares of Fortune Pharma Ltd. of Re. 1 each, without any consideration to the shareholders of Fortune India Ltd. For that purpose following points are to be considered:

(a) Transfer of Liabilities & Assets at Book value.

(b) Estimated Profit for the year 2009-10 is ₹ 11,400 Lakh for Fortune India Ltd. & ₹ 1,470 lakhs for Fortune Pharma Ltd.

(c) Estimated Market Price of Fortune Pharma Ltd. is ₹ 24.50 per share.

(d) Average P/E Ratio of FMCG sector is 42 & Pharma sector is 25, which is to be expected for both the companies.

Calculate:

1. The Ratio in which shares of Fortune Pharma are to be issued to the shareholders of Fortune India Ltd.
2. Expected Market price of Fortune India (FMCG) Ltd.
3. Book Value per share of both the Companies immediately after Demerger.

Solution — ICAI

22. Share holders' funds **(₹ Lakhs)**

Particulars	Fortune India Ltd.	Fortune Pharma Ltd.	Fortune India (FMCG) Ltd.
Assets	70,000	25,100	44,900
Outside liabilities	25,000	4,100	20,900
Net worth	45,000	21,000	24,000

1. Calculation of Shares of Fortune Pharma Ltd. to be issued to shareholders of Fortune India Ltd.

	Fortune Pharma Ltd.
Estimated Profit (₹ in lakhs)	1,470
Estimated market price (₹)	24.5
Estimated P/E	25
Estimated EPS (₹)	0.98
No. of shares lakhs	1,500

Hence, Ratio is 1 share of Fortune Pharma Ltd. for 2 shares of Fortune India Ltd.

OR for 0.50 share of Fortune Pharma Ltd. for 1 share of Fortune India Ltd.

2. Expected market price of Fortune India (FMCG) Ltd.

	Fortune India (FMCG) Ltd.
Estimated Profit (₹ in lakhs)	11,400
No. of equity shares (₹ in lakhs)	3,000
Estimated EPS (₹)	3.8
Estimated P/E	42
Estimated market price (₹)	159.60

3. Book value per share

	Fortune Pharma Ltd.	Fortune India (FMCG) Ltd.
Net worth (₹in lakhs)	21,000	24,000
No. of shares (₹ in lakhs)	1,500	3,000
Book value of shares	₹ 14	₹ 8

Practical Q23

23. H Ltd. agrees to buy over the business of B Ltd. effective 1st April, 2012.The summarized Balance Sheets of H Ltd. and B Ltd. as on 31st March 2012 are as follows:

Balance sheet as at 31st March, 2012 (In Crores of Rupees)

Liabilities:	H. Ltd	B. Ltd.
Paid up Share Capital		
-Equity Shares of ₹100 each	350.00	
-Equity Shares of ₹10 each		6.50
Reserve & Surplus	950.00	25.00
Total	**1,300.00**	**31.50**
Assets:		
Net Fixed Assets	220.00	0.50
Net Current Assets	1,020.00	29.00
Deferred Tax Assets	60.00	2.00
Total	**1,300.00**	**31.50**

H Ltd. proposes to buy out B Ltd. and the following information is provided to you as part of the scheme of buying:

(a) The weighted average post tax maintainable profits of H Ltd. and B Ltd. for the last 4 years are ₹ 300 crores and ₹ 10 crores respectively.

(b) Both the companies envisage a capitalization rate of 8%.

(c) H Ltd. has a contingent liability of ₹ 300 crores as on 31st March, 2012.

(d) H Ltd. to issue shares of ₹ 100 each to the shareholders of B Ltd. in terms of the exchange ratio as arrived on a Fair Value basis. (Please consider weights of 1 and 3 for the value of shares arrived on Net Asset basis and Earnings capitalization method respectively for both H Ltd. and B Ltd.)

You are required to arrive at the value of the shares of both H Ltd. and B Ltd. under:

(i) Net Asset Value Method

(ii) Earnings Capitalisation Method

(iii) Exchange ratio of shares of H Ltd. to be issued to the shareholders of B Ltd. on a Fair value basis (taking into consideration the assumption mentioned in point 4 above.)

Solution — ICAI

23. (i) Net asset value

H Ltd.	$\frac{₹\ 1300\ \text{Crores} - ₹\ 300\ \text{Crores}}{3.50\ \text{Crores}} = ₹\ 285.71$
B Ltd.	$\frac{₹\ 31.50\ \text{Crores}}{0.65\ \text{Crores}} = ₹\ 48.46$

(ii) Earning capitalization value

H Ltd.	$\frac{₹\ 300\ \text{Crores} / 0.08}{3.50\ \text{Crores}} = ₹\ 1071.43^*$
B Ltd.	$\frac{₹\ 10\ \text{Crores} / 0.08}{0.65\ \text{Crores}} = ₹\ 192.31$

* Alternatively, Contingent Liability can also be deducted from this Valuation.

(iii) Fair value

H Ltd.	$\frac{₹\ 285.71 \times 1 + ₹\ 1071.43 \times 3}{4} = ₹\ 875$
B Ltd.	$\frac{₹\ 48.46 \times 1 + ₹\ 192.31 \times 3}{4} = ₹\ 156.3475$

Exchange ratio ₹156.3475/ ₹875 = 0.1787

H Ltd should issue its 0.1787 share for each share of B Ltd.

Note: In above solution it has been assumed that the contingent liability will materialize at its full amount.

Practical Q24

24. Reliable Industries Ltd. (RIL) is considering a takeover of Sunflower Industries Ltd. (SIL). The particulars of 2 companies are given below:

Particulars	Reliable Industries Ltd	Sunflower Industries Ltd.
Earnings After Tax (EAT)	₹ 20,00,000	₹ 10,00,000
Equity shares O/s	10,00,000	10,00,000
Earnings per share (EPS)	2	1
PE Ratio (Times)	10	5

Required:

(i) What is the market value of each Company before merger?

(ii) Assume that the management of RIL estimates that the shareholders of SIL will accept an offer of one share of RIL for four shares of SIL. If there are no synergic effects, what is the market value of the Post-merger RIL? What is the new price per share? Are the shareholders of RIL better or worse off than they were before the merger?

(iii) Due to synergic effects, the management of RIL estimates that the earnings will increase by 20%. What are the new post-merger EPS and Price per share? Will the shareholders be better off or worse off than before the merger?

Solution — ICAI

24. (i) Market value of Companies before Merger

Particulars	*RIL*	*SIL*
EPS	₹ 2	Re.1
P/E Ratio	10	5
Market Price Per Share	₹ 20	₹ 5
Equity Shares	10,00,000	10,00,000
Total Market Value	2,00,00,000	50,00,000

(ii) Post Merger Effects on RIL

	₹
Post-merger earnings	30,00,000
Exchange Ratio (1:4)	
No. of equity shares o/s (10,00,000 + 2,50,000)	12,50,000
EPS: 30,00,000/12,50,000	2.4
PE Ratio	10
Market Value 10 x 2.4	24
Total Value (12,50,000 x 24)	3,00,00,000

Gains From Merger:	₹
Post-Merger Market Value of the Firm	3,00,00,000
Less: Pre-Merger Market Value	
RIL 2,00,00,000	
SIL 50,00,000	2,50,00,000
Total gains from Merger	50,00,000

Apportionment of Gains between the Shareholders:

Particulars	**RIL (₹)**	**SIL (₹)**
Post-Merger Market Value:		
10,00,000 x 24	2,40,00,000	--
2,50,000 x 24	-	60,00,000
Less: Pre-Merger Market Value	2,00,00,000	50,00,000
Gains from Merger:	40,00,000	10,00,000

Thus, the shareholders of both the companies (RIL + SIL) are better off than before

(iii) Post-Merger Earnings:

Increase in Earnings by 20%

New Earnings: ₹ 30,00,000 x (1+0.20)	₹ 36,00,000
No. of equity shares outstanding:	12,50,000
EPS (₹ 36,00,000/12,50,000)	₹ 2.88
PE Ratio	10
Market Price Per Share: = ₹2.88 x 10	₹ 28.80

∴ Shareholders will be better-off than before the merger situation.

Practical Q25

25. AFC Ltd. wishes to acquire BCD Ltd. The shares issued by the two companies are 10,00,000

and 5,00,000 respectively:

(i) Calculate the increase in the total value of BCD Ltd. resulting from the acquisition on the basis of the following conditions:

Current expected growth rate of BCD Ltd.		7%
Expected growth rate under control of AFC Ltd., (without any additional capital investment and without any change in risk of operations)		8%
Current Market price per share of AFC Ltd.	₹	100
Current Market price per share of BCD Ltd.	₹	20
Expected Dividend per share of BCD Ltd.	₹	0.60

(ii) On the basis of aforesaid conditions calculate the gain or loss to shareholders of both the companies, if AFC Ltd. were to offer one of its shares for every four shares of BCD Ltd.

(iii) Calculate the gain to the shareholders of both the Companies, if AFC Ltd. pays ₹22 for each share of BCD Ltd., assuming the P/E Ratio of AFC Ltd. does not change after the merger. EPS of AFC Ltd. is ₹8 and that of BCD is ₹2.50. It is assumed that AFC Ltd. invests its cash to earn 10%.

Solution — ICAI

25. (i) For BCD Ltd., before acquisition

The cost of capital of BCD Ltd. may be calculated by using the following formula:

$$\frac{\text{Dividend}}{\text{Price}} + \text{Growth}\%$$

Cost of Capital i.e., Ke = (0.60/20) + 0.07 = 0.10

After acquisition g (i.e. growth) becomes 0.08

Therefore, price per share after acquisition = 0.60/(0.10-0.08) = ₹ 30

The increase in value therefore is = ₹(30-20) x 5,00,000 = ₹ 50,00,000

(ii) To shareholders of BCD Ltd. the immediate gain is ₹ 100 – ₹ 20 x 4 = ₹ 20 per share

The gain can be higher if price of shares of AFC Ltd. rise following merger which they should undertake.

To AFC Ltd. shareholders	(In ₹ lakhs)
Value of Company now	1,000
Value of BCD Ltd.	150
	1,150
No. of shares	11.25
∴ Value per share	1150/11.25 = ₹ 102.22

Gain to shareholders of BCD Ltd. = ₹ 102.22 – ₹(4 x 20) = ₹ 22.22

Gain to shareholders of AFC Ltd. = ₹ 102.22 – ₹ 100.00 = ₹ 2.22

(iii) Gain to shareholders of AFC Ltd:-

Earnings of BCD Ltd. (5,00,000 x 2.50)	₹ 12,50,000/-
Less: Loss of earning in cash (5,00,000 x ₹ 22 x 0.10)	₹ 11,00,000/-
Net Earning	₹ 1,50,000/-
Number of shares	10,00,000
Net increase in earning per share	0.15

P/E ratio of AFC Ltd. = 100/8 = 12.50

Therefore, Gain per share of shareholders of AFC Ltd. = 0.15x12.50 = ₹ 1.88

Gain to the shareholders of BCD Ltd. ₹ (22-20) = ₹ 2/- per share

Alternatively, it can also be computed as follows:

Post-Merger Earnings (10,00,000 x ₹ 8 + 5,00,000 x ₹ 2.5 – 11,00,000)	₹ 81,50,000
EPS after Merger $\left(\frac{81,50,000}{10,00,000}\right)$	₹ 8.15
PE Ratio	12.50
Post Merger Price of Share (₹ 8.15 x 12.50)	₹ 101.875
Less: Price before merger	₹ 100.00
	₹ 1.875
Say	₹ 1.88

Practical Q26

26. AB Ltd., is planning to acquire and absorb the running business of XY Ltd. The valuation is to be based on the recommendation of merchant bankers and the consideration is to be discharged in the form of equity shares to be issued by AB Ltd. As on 31.3.2006, the paid up capital of AB Ltd. consists of 80 lakhs shares of ₹10 each. The highest and the lowest market quotation during the last 6 months were ₹570 and ₹430. For the purpose of the exchange, the price per share is to be reckoned as the average of the highest and lowest market price during the last 6 months ended on 31.3.06.

XY Ltd.'s Balance Sheet as at 31.3.2006 is summarised below:

	₹ lakhs
Sources	
Share Capital	
20 lakhs equity shares of ₹10 each fully paid	200
10 lakhs equity shares of ₹10 each, ₹5 paid	50
Loans	100
Total	350

Uses	
Fixed Assets (Net)	150
Net Current Assets	200
	350

An independent firm of merchant bankers engaged for the negotiation, have produced the following estimates of cash flows from the business of XY Ltd.:

Year ended	By way of	₹ lakhs
31.3.07	after tax earnings for equity	105
31.3.08	do	120
31.3.09	Do	125
31.3.10	Do	120
31.3.11	Do	100
	Terminal Value estimate	200

It is the recommendation of the merchant banker that the business of XY Ltd. may be valued on the basis of the average of (i) Aggregate of discounted cash flows at 8% and (ii) Net assets value. Present value factors at 8% for years

1-5: 0.93 0.86 0.79 0.74 0.68

You are required to:

(a) Calculate the total value of the business of XY Ltd.

(b) The number of shares to be issued by AB Ltd.; and

(c) The basis of allocation of the shares among the shareholders of XY Ltd.

Solution — ICAI

26. Price/share of AB Ltd. for determination of number of shares to be issued

= (₹ 570 + ₹ 430)/2 = ₹ 500

Value of XY Ltd based on future cash flow capitalization (105x0.93)+(120x0.86)+(125x0.79)+(120x0.74)x(300x0.68)	₹ lakhs	592.40
Value of XY Ltd based on net assets	₹ lakhs	250.00
Average value (592.40+250)/2		421.20
No. of shares in AB Ltd to be issued ₹ 4,21,20,000/500	Nos.	84240
Basis of allocation of shares		
Fully paid equivalent shares in XY Ltd. (20+5) lakhs		2500000
Distribution to fully paid shareholders 84240x20/25		67392
Distribution to partly paid shareholders 84240-67392		16848

Practical Q27

27. R Ltd. and S Ltd. are companies that operate in the same industry. The financial statements of both the companies for the current financial year are as follows:

Balance Sheet

Particulars	R. Ltd. (₹)	S. Ltd (₹)
Equity & Liabilities		
Shareholders Fund		
Equity Capital (₹ 10 each)	20,00,000	16,00,000
Retained earnings	4,00,000	-
Non-current Liabilities		
16% Long term Debt	10,00,000	6,00,000
Current Liabilities	14,00,000	8,00,000
Total	**48,00,000**	**30,00,000**
Assets		
Non-current Assets	20,00,000	10,00,000
Current Assets	28,00,000	20,00,000
Total	**48,00,000**	**30,00,000**

Income Statement

	Particulars	R. Ltd. (₹)	S. Ltd. (₹)
A.	Net Sales	69,00,000	34,00,000
B.	Cost of Goods sold	55,20,000	27,20,000
C.	**Gross Profit (A-B)**	13,80,000	6,80,00
D.	Operating Expenses	4,00,000	2,00,000
E.	Interest	1,60,000	96,000
F.	**Earnings before taxes [C-(D+E)]**	8,20,000	3,84,000
G.	Taxes @ 35%	2,87,000	1,34,400
H.	Earnings After Tax (EAT)	5,33,000	2,49,600

Additional Information:

No. of equity shares	2,00,000	1,60,000
Dividend payment Ratio (D/P)	20%	30%
Market price per share	₹ 50	₹ 20

Assume that both companies are in the process of negotiating a merger through exchange of Equity shares:

You are required to:

(i) Decompose the share price of both the companies into EPS & P/E components. Also segregate their EPS figures into Return On Equity (ROE) and Book Value/Intrinsic Value per share components.

(ii) Estimate future EPS growth rates for both the companies.

(iii) Based on expected operating synergies, R Ltd. estimated that the intrinsic value of S Ltd. Equity share would be ₹ 25 per share on its acquisition. You are required to develop a range of justifiable Equity Share Exchange ratios that can be offered by R Ltd. to the shareholders of S Ltd. Based on your analysis on parts (i) and (ii), would

you expect the negotiated terms to be closer to the upper or the lower exchange ratio limits and why?

Solution — ICAI

27. (i) Determination of EPS, P/E Ratio, ROE and BVPS of R Ltd. & S Ltd.

	R Ltd.	S Ltd.
EAT (₹)	5,33,000	2,49,600
N	200000	160000
EPS (EAT÷N)	2.665	1.56
Market Price Per Share	50	20
PE Ratio (MPS/EPS)	18.76	12.82
Equity Fund (Equity Value)	2400000	1600000
BVPS (Equity Value ÷ N)	12	10
ROE (EAT÷ EF) or	0.2221	0.156
ROE (EAT ÷ EF) x 100	22.21%	15.60%

(ii) Determination of Growth Rate of EPS of R Ltd.& S Ltd.

	R Ltd.	S Ltd.
Retention Ratio (1-D/P Ratio)	0.80	0.70
Growth Rate (ROE x Retention Ratio) or	0.1777	0.1092
Growth Rate (ROE x Retention Ratio) x 100	17.77%	10.92%

(iii) Justifiable equity share exchange ratio

(a) Market Price Based = MPS_S/MPS_R = ₹ 20/ ₹ 50 = 0.40:1 (lower limit)

(b) Intrinsic Value Based = ₹ 25/ ₹ 50 = 0.50:1 (max. limit)

Since R Ltd. has higher EPS, PE, ROE and higher growth expectations the negotiated term would be expected to be closer to the lower limit, based on existing share price.

Practical Q28

28. BA Ltd. and DA Ltd. both the companies operate in the same industry. The Financial statements of both the companies for the current financial year are as follows:

Balance Sheet

Particulars	BA Ltd. (₹)	DA Ltd. (₹)
Current Assets	14,00,000	10,00,000
Fixed Assets (Net)	10,00,000	5,00,000
Total (₹)	**24,00,000**	**15,00,000**
Equity capital (₹10 each)	10,00,000	8,00,000
Retained earnings	2,00,000	--
14% long-term debt	5,00,000	3,00,00
Current liabilities	7,00,000	4,00,000
Total (₹)	**24,00,000**	**15,00,000**

Income Statement

	BA Ltd. (₹)	DA Ltd. (₹)
Net Sales	34,50,000	17,00,000
Cost of Goods sold	27,60,000	13,60,000
Gross profit	6,90,000	3,40,000
Operating expenses	2,00,000	1,00,000
Interest	70,000	42,000
Earnings before taxes	4,20,000	1,98,00
Taxes @ 50%	2,10,000	99,000
Earnings after taxes (EAT)	**2,10,000**	**99,000**
Additional Information :		
No. of Equity shares	1,00,000	80,000
Dividend payment ratio (D/P)	40%	60%
Market price per share	₹40	₹15

Assume that both companies are in the process of negotiating a merger through an exchange of equity shares. You have been asked to assist in establishing equitable exchange terms and are required to:

(i) Decompose the share price of both the companies into EPS and P/E components; and also segregate their EPS figures into Return on Equity (ROE) and book value/intrinsic value per share components.

(ii) Estimate future EPS growth rates for each company.

(iii) Based on expected operating synergies BA Ltd. estimates that the intrinsic value of DA's equity share would be ₹20 per share on its acquisition. You are required to develop a range of justifiable equity share exchange ratios that can be offered by BA Ltd. to the shareholders of DA Ltd. Based on your analysis in part (i) and (ii), would you expect the negotiated terms to be closer to the upper, or the lower exchange ratio limits and why?

(iv) Calculate the post-merger EPS based on an exchange ratio of 0.4: 1 being offered by BA Ltd. and indicate the immediate EPS accretion or dilution, if any, that will occur for each group of shareholders.

(v) Based on a 0.4: 1 exchange ratio and assuming that BA Ltd.'s pre-merger P/E ratio will continue after the merger, estimate the post-merger market price. Also show the resulting accretion or dilution in pre-merger market prices.

Solution — ICAI

28. Market price per share (MPS) = EPS X P/E ratio or P/E ratio = MPS/EPS

(i) Determination of EPS, P/E ratio, ROE and BVPS of BA Ltd. and DA Ltd.

		BA Ltd.	DA Ltd.
Earnings After Tax	(EAT)	₹ 2,10,000	₹ 99,000
No. of Shares	(N)	100000	80000
EPS	(EAT/N)	₹ 2.10	₹ 1.2375
Market price per share	(MPS)	40	15
P/E Ratio	(MPS/EPS)	19.05	12.12
Equity Funds	(EF)	₹ 12,00,000	₹ 8,00,000
BVPS	(EF/N)	12	10
ROE	(EAT/EF) × 100	17.50%	12.37%

(ii) Estimation of growth rates in EPS for BA Ltd. and DA Ltd.

Retention Ratio	(1-D/P ratio)	0.6	0.4
Growth Rate	(ROE × Retention Ratio)	10.50%	4.95%

(iii) Justifiable equity shares exchange ratio

(a) Intrinsic value based = ₹20 / ₹40 = 0.5:1 (upper limit)

(b) Market price based = MPS_{DA}/MPS_{BA} = ₹15 / ₹40 =0.375:1(lower limit)

Since, BA Ltd. has a higher EPS, ROE, P/E ratio and even higher EPS growth expectations, the negotiable terms would be expected to be closer to the lower limit, based on the existing share prices.

(iv) Calculation of Post merger EPS and its effects

Particulars			BA Ltd.	DA Ltd.	Combined
EAT	(₹)	(i)	2,10,000	99,000	3,09,000
Share outstanding		(ii)	100000	80000	132000*
EPS	(₹)	(i) / (ii)	2.1	1.2375	2.341
EPS Accretion (Dilution)	(Re.)		0.241	(0.301***)	

(v) Estimation of Post merger Market price and other effects

Particulars			BA Ltd.	DA Ltd.	Combined
EPS	(₹)	(i)	2.1	1.2375	2.341
P/E Ratio		(ii)	19.05	12.12	19.05
MPS	(₹)	(i) / (ii)	40	15	44.6
MPS Accretion	(₹)		4.6	2.84***	

* Shares outstanding (combined)	= 100000 shares + (0.40 × 80000) = 132000 shares	
** EPS claim per old share	= ₹ 2.34 × 0.4	₹ 0.936
EPS dilution	= ₹ 1.2375 – ₹ 0.936	₹ 0.3015
***S claim per old share	(₹ 44.60 × 0.4)	₹ 17.84
Less: MPS per old share		₹ 15.00
		₹ 2.84

Practical Q29

29. During the audit of the Weak Bank (W), RBI has suggested that the Bank should either merge with another bank or may close down. Strong Bank (S) has submitted a proposal of merger of Weak Bank with itself. The *relevant information and Balance Sheets of both the companies are as under:*

Particulars	Weak Bank (W)	Strong Bank (S)	Assigned Weights (%)
Gross NPA (%)	40	5	30
Capital Adequacy Ratio (CAR)	5	16	28
Total Capital/ Risk Weight Asset			
Market price per Share (MPS)	12	96	32
Book value			10
Trading on Stock Exchange	Irregular	Frequent	

Balance Sheet **(₹ in Lakhs)**

Particulars	Weak Bank (W)	Strong Bank (S)
Paid up Share Capital (₹ 10 per share)	150	500
Reserves & Surplus	80	5,500
Deposits	4,000	44,000
Other Liabilities	890	2,500
Total Liabilities	**5,120**	**52,500**
Cash in Hand & with RBI	400	2,500
Balance with Other Banks	-	2,000
Investments	1,100	19,000
Advances	3,500	27,000
Other Assets	70	2,000
Preliminary Expenses	50	-
Total Assets	**5,120**	**52,500**

You are required to

(a) Calculate Swap ratio based on the above weights:

(b) Ascertain the number of Shares to be issued to Weak Bank;

(c) Prepare Balance Sheet after merger; and

(d) Calculate CAR and Gross NPA of Strong Bank after merger.

Solution — ICAI

29. (a) Swap Ratio

Gross NPA	5:40	5/40 x 30%	0.0375
CAR	5:16	5/16 x 28%	0.0875
Market Price	12:96	12/96 x 32%	0.0400
Book Value Per Share	12:120	12/120x 10%	0.0100
			0.1750

Thus, for every share of Weak Bank, 0.1750 share of Strong Bank shall be issued.

Calculation of Book Value Per Share

Particulars	Weak Bank (W)	Strong Bank (S)
Share Capital (₹ Lakhs)	150	500
Reserves & Surplus (₹ Lakhs)	80	5,500
	230	6,000
Less: Preliminary Expenses (₹ Lakhs)	50	--
Net Worth or Book Value (₹ Lakhs)	180	6,000
No. of Outstanding Shares (Lakhs)	15	50
Book Value Per Share (₹)	12	120

(b) No. of equity shares to be issued:

$$\frac{150}{10} \times 0.1750 = 2.625 \text{ lakh shares}$$

(c) Balance Sheet after Merger

Calculation of Capital Reserve

Book Value of Shares	₹ 180.00 lac
Less: Value of Shares issued	₹ 26.25 lac
Capital Reserve	₹ 153.75 lac

Balance Sheet

	₹ lac		₹ lac
Paid up Share Capital	526.25	Cash in Hand & RBI	2900.00
Reserves & Surplus	5500.00	Balance with other banks	2000.00
Capital Reserve	153.75	Investment	20100.00
Deposits	48000.00	Advances	30500.00
Other Liabilities	3390.00	Other Assets	2070.00
	57570.00		**57570.00**

(d) Calculation CAR & Gross NPA % of Bank 'S' after merger

$$CAR / CRWAR = \frac{\text{Total Capital}}{\text{Risky Weighted Assets}}$$

	Weak Bank	Strong Bank	Merged
Capital Adequacy Ratio (CAR)	5%	16%	
Total Capital	₹ 180 lac	₹ 6000 lac	₹ 6180 lac
Risky Weighted Assets	₹ 3600 lac	₹ 37500 lac	₹ 41100 lac

$$CAR = \frac{6180}{41100} \times 100 = 15.04\%$$

$$\text{GNPA Ratio} = \frac{\text{Gross NPA}}{\text{Gross Advances}} \times 100$$

	Weak Bank	Strong Bank	Merged
GNPA (Given)	0.40	0.05	
	$0.40 = \frac{GNPA_R}{₹\ 3500\ lac}$	$0.05 = \frac{GNPA_S}{₹\ 27000\ lac}$	
Gross NPA	₹ 1400 lac	₹ 1350 lac	₹ 2750 lac

Practical Q30

30. M/s Tiger Ltd. wants to acquire M/s. Leopard Ltd. The balance sheet of Leopard Ltd. as on 31st March, 2012 is as follows:

Liabilities	₹	Assets	₹
Equity Capital (70,000 shares)	7,00,000	Cash	50,000
Retained earnings	3,00,000	Debtors	70,000
12% Debentures	3,00,000	Inventories	2,00,000
Creditors and other liabilities	3,20,000	Plants & Eqpt.	13,00,000
	16,20,000		16,20,000

Additional Information:

(i) Shareholders of Leopard Ltd. will get one share in Tiger Ltd. for every two shares. External liabilities are expected to be settled at ₹ 5,00,000. Shares of Tiger Ltd. would be issued at its current price of ₹ 15 per share. Debenture holders will get 13% convertible debentures in the purchasing company for the same amount. Debtors and inventories are expected to realize ₹ 2,00,000.

(ii) Tiger Ltd. has decided to operate the business of Leopard Ltd. as a separate division. The division is likely to give cash flows (after tax) to the extent of ₹ 5,00,000 per year

for 6 years. Tiger Ltd. has planned that, after 6 years, this division would be demerged and disposed of for ₹ 2,00,000.

(iii) The company's cost of capital is 16%.

Make a report to the Board of the company advising them about the financial feasibility of this acquisition.

Net present values for 16% for ₹ 1 are as follows:

Years	1	2	3	4	5	6
PV	0.862	0.743	0.641	0.552	0.476	0.410

Solution — ICAI

30. Calculation of Purchase Consideration

	₹
Issue of Share 35000 x ₹15	5,25,000
External Liabilities settled	5,00,000
13% Debentures	3,00,000
	13,25,000
Less: Realization of Debtors and Inventories	2,00,000
Cash	50,000
	10,75,000

Net Present Value = PV of Cash Inflow + PV of Demerger of Leopard Ltd. – Cash Outflow

= ₹ 5,00,000 PVAF(16%,6) + ₹ 2,00,000 PVF(16%, 6) – ₹ 10,75,000

= ₹ 5,00,000 x 3.684 + ₹ 2,00,000 x 0.410 – ₹ 10,75,000

= ₹ 18,42,000 + ₹ 82,000 – ₹ 10,75,000

= ₹ 8,49,000

Since NPV of the decision is positive it is advantageous to acquire Leopard Ltd.

Practical Q31

31. The equity shares of XYZ Ltd. are currently being traded at ₹ 24 per share in the market. XYZ Ltd. has total 10,00,000 equity shares outstanding in number; and promoters' equity holding in the company is 40%.

PQR Ltd. wishes to acquire XYZ Ltd. because of likely synergies. The estimated present value of these synergies is ₹ 80,00,000.

Further PQR feels that management of XYZ Ltd. has been over paid. With better motivation, lower salaries and fewer perks for the top management, will lead to savings of ₹ 4,00,000 p.a. Top management with their families are promoters of XYZ Ltd. Present value of these savings would add ₹ 30,00,000 in value to the acquisition.

Following additional information is available regarding PQR Ltd.:

Earnings per share	: ₹ 4
Total number of equity shares outstanding	: 15,00,000
Market price of equity share	: ₹ 40

Required:

(i) What is the maximum price per equity share which PQR Ltd. can offer to pay for XYZ Ltd.?

(ii) What is the minimum price per equity share at which the management of XYZ Ltd. will be willing to offer their controlling interest?

Solution — ICAI

31. (i) Calculation of maximum price per share at which PQR Ltd. can offer to pay for XYZ Ltd.'s share

Market Value (10,00,000 x ₹ 24)	₹ 2,40,00,000
Synergy Gain	₹ 80,00,000
Saving of Overpayment	₹ 30,00,000
	₹ 3,50,00,000
Maximum Price (₹ 3,50,00,000/10,00,000)	₹ 35

Alternatively, it can also be computed as follows:

Let ER be the swap ratio then,

$$40 = \frac{24 \times 10,00,000 + 40 \times 15,00,000 + 80,00,000 + 30,00,000}{15,00,000 + 10,00,000 \times ER}$$

ER = 0.875

$$MP = PE \times EPS \times ER = \frac{40}{4} \times ₹\ 4 \times 0.875 = ₹\ 35$$

(ii) Calculation of minimum price per share at which the management of XYZ Ltd.'s will be willing to offer their controlling interest

Value of XYZ Ltd.'s Management Holding (40% of 10,00,000 x ₹ 24)	₹ 96,00,000
Add: PV of loss of remuneration to top management	₹ 30,00,000
	₹ 1,26,00,000
No. of Shares	4,00,000
Minimum Price (₹ 1,26,00,000/4,00,000)	₹ 31.50

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